

PCA9430

NFC charging power receiver with linear battery charger

Rev. 3.0 — 2 December 2020
652730

Product data sheet
COMPANY PROPRIETARY

1 General description

The PCA9430 is an integrated low-power wireless charging receiver, which features antenna tuning capability. It targets NFC-based wireless charging applications, where power can be transferred through the inductive coupling between the two loop antennas for low power charging.

The device operates with AC inputs, and converts into an internal voltage, then feeds into the linear battery charger block. The receiver includes an integrated full bridge rectifier and internal regulation circuitry suitable for single cell li-ion battery charging. It also features a 5-bit on-chip DAC which can be used for antenna tuning. The integrated 10-bit ADC monitors various voltages, currents, on-chip temperature and external NTC for battery temperature. Overall limited external components are used to reduce the total solution size and cost.

Other features such as FM+ I²C-bus interface, chip enable, interrupt signal, etc. are also provided.

The PCA9430 is offered in 2.2 x 2.8 x 0.50 mm XQFN16 package.

2 Features and benefits

- Supports NFC forum wireless charger
 - 200 mW with internal rectifier
 - 1 W with external rectifier
- Automatic overvoltage protection of the antenna inputs
- Integrated 1-cell Li-ion linear battery charger
- Auxiliary 50 mA VPWR low-dropout regulator (LDO) for dead battery supply
- 10-bit multiple channel ADC
- 5-bit DAC for antenna tuning
- External temperature sensing analog interface for NTC
- 1 MHz Fast Mode Plus I²C-bus slave interface
- Protection circuitry
 - Over-Temperature Protection
 - VRECT Overvoltage Protection
 - VRECT Undervoltage Protection
- ESD protection
 - HBM ANSI/ESDA/JEDEC JS-001 Class 1C exceeds 1.5 kV on all pins
 - CDM ANSI/ESDA/JEDEC JS-002 exceeds 500 V
- Operating ambient temperature -40 °C to +85 °C



3 Applications

- Wearables and Healthcare
- Smart Home
- Stylus pens
- Smart IoT products
- Portable devices

4 Ordering information

Table 1. Ordering information

Type number	Package			
	Temperature range	Name	Description	Version
PCA9430HK	-40 °C to +85 °C	XQFN16	plastic, extremely thin quad flat package; no leads; 16 terminals; body 2.2 x 2.8 x 0.50 mm	SOT1956-1

4.1 Ordering options

Table 2. Ordering options

Type number	Orderable part number	Package	Packing method	Minimum order quantity	Temperature
PCA9430HK	PCA9430HK	XQFN16	reel dry pack, SMD, 7" Q1 standard product orientation	4000	T _{amb} = -40 °C to +85 °C

5 Marking

Table 3. Marking

Line	Content	Description
1	Pin 1 dot	Pin 1 dot
	30	Product identification
2	XXX	3 digit lot number
3	YWW	Y: Last digits of year code of assembly, WW: week code of assembly

6 Functional diagram

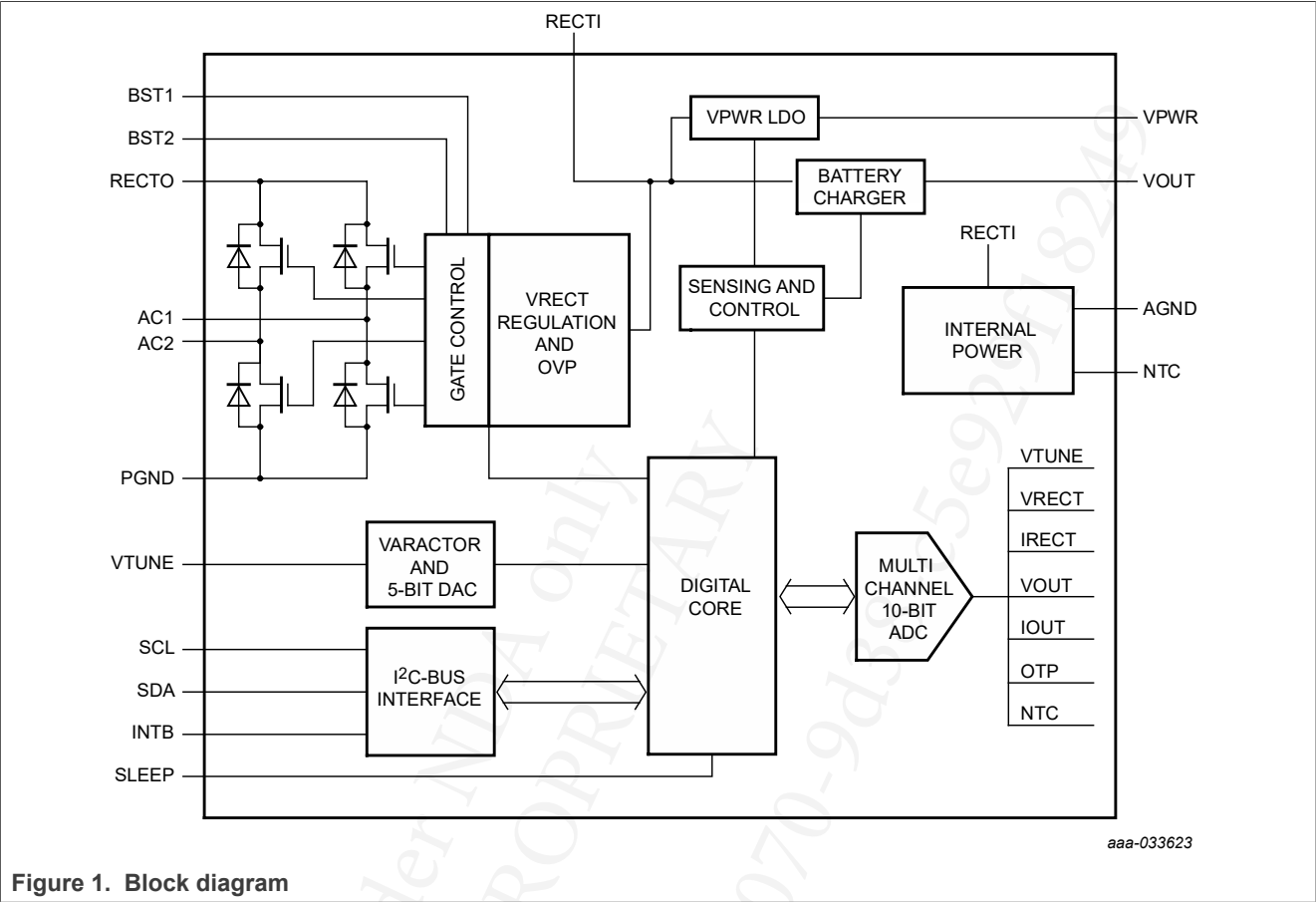


Figure 1. Block diagram

7 Pinning information

7.1 Pinning

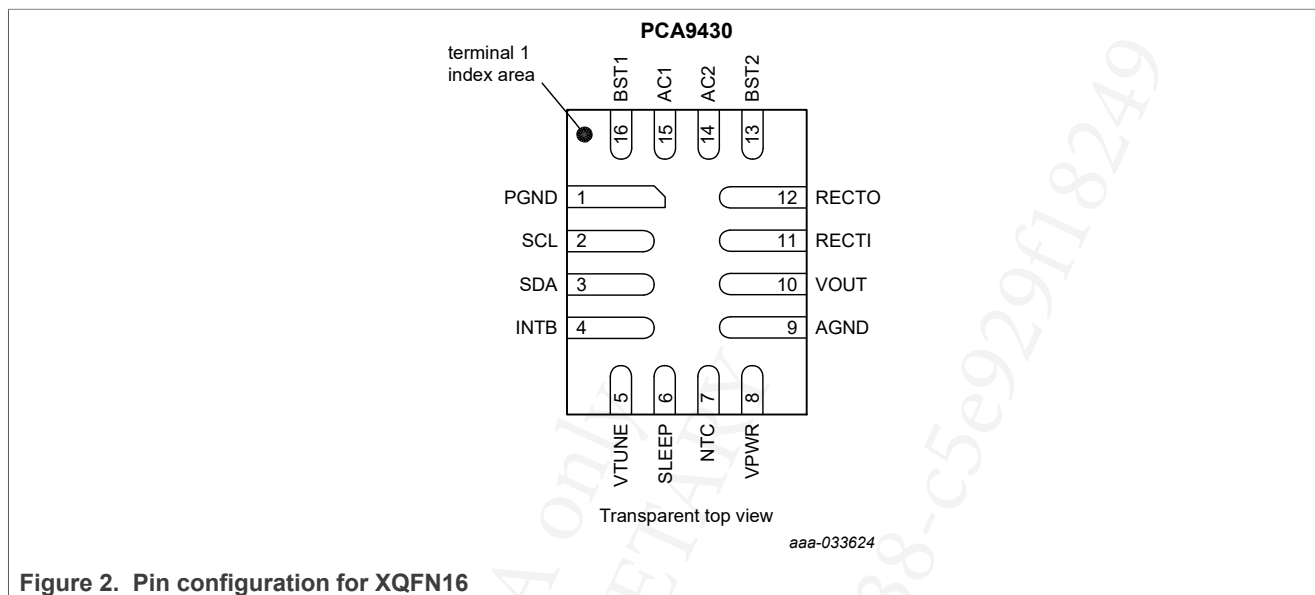


Figure 2. Pin configuration for XQFN16

7.2 Pin description

Table 4. Pin description

Symbol	Pin	Description
PGND	1	Power ground pin
SCL	2	I ² C Clock input
SDA	3	I ² C Data input/output
INTB	4	open drain output; I ² C interrupt and chip alert
VTUNE	5	DAC output for varactor capacitor antenna tuning adjustment
SLEEP	6	SLEEP mode enable pin. The pin has 1 MΩ pull down resistor integrated
NTC	7	Battery temperature sensing pin. The external thermistor is connected between NTC pin and ground
VPWR	8	Auxiliary LDO power output pin for dead battery supply
AGND	9	Analog ground pin
VOUT	10	Linear charger output pin
RECTI	11	Rectifier voltage input pin to internal circuit block. It is mandatory to connect this pin to RECTO in PCB level after the capacitance when internal rectifier is used
RECTO	12	Rectifier voltage output pin
BST2	13	Bootstrap capacitor pin 2 for rectifier high side MOSFET
AC2	14	AC input power pin 2, connected to resonator AC power output
AC1	15	AC input power pin 1, connected to resonator AC power output
BST1	16	Bootstrap capacitor pin 1 for rectifier high side MOSFET

8 Functional description

8.1 General function descriptions

PCA9430 is a low-power wireless charging receiver with linear battery charger function. It can be controlled via I²C-bus interface by a host microcontroller (MCU) or application processor (AP). The device operates with AC input for less than 200 mW, through an internal full bridge rectifier, and converts into the internal linear battery charger to charge a single-cell li-ion battery. External rectifier is required when needed application power is higher than 200 mW

The device features the capability to tune the antenna varactor via an on-chip 5-bit DAC, while a 10-bit on-chip ADC monitors multiple system parameters of the chip. In addition, an NTC pin is provided to monitor an external NTC thermistor and resistor ladder for temperature monitoring on system PCB board. Meanwhile, comprehensive chip protection, including input overvoltage protection, is offered for system level safety.

8.2 Sleep mode

The sleep mode is designed for low power consumption. If there is no REG_HIGH or overvoltage protection fault, PCA9430 enters sleep mode when system MCU pulls SLEEP pin to HIGH. If REG_HIGH or overvoltage protection fault occurs, PCA9430 will not enter sleep mode.

While in sleep mode, the system MCU can wake up PCA9430 by driving SLEEP pin to LOW. The device enters ECO_Mode and the system MCU configures PCA9430 for charging. If the REG_HIGH or overvoltage protection fault alerts in sleep mode, the device wakes up and enters ECO_Mode, so that the interrupt can be sent to system MCU to negotiate for lower power if necessary.

The pin has an internal integrated 1 M Ω pull down resistor, which ensures the linear charger is active when the system battery is missing or dead.

8.3 Power on from AC1 and AC2

PCA9430 is powered by an RF coil through AC1 and AC2 pins with an internal rectifier. [Figure 3](#) shows the power on flow chart for PCA9430.

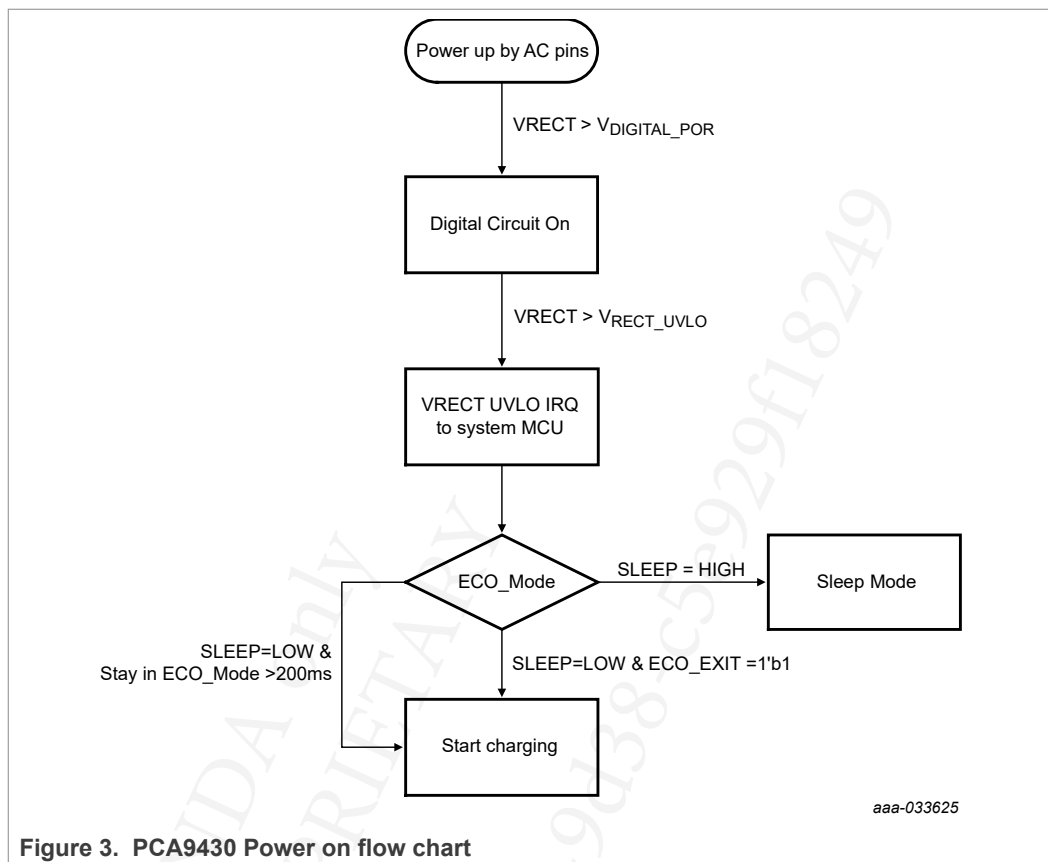


Figure 3. PCA9430 Power on flow chart

When VRECT voltage exceeds V_{RECT_UVLO} , the device issues VRECT RDY interrupt to notify the system MCU that power is ready in PCA9430 and enters ECO_Mode. In this mode, the system MCU can configure PCA9430 based on system requirements. When the configuration is done, the system MCU can write ECO_EXIT=1'b1 to start charging directly. If the battery is full and charging is not needed, PCA9430 can enter sleep mode by setting SLEEP pin to HIGH. If the system MCU does not have any response in ECO_Mode, such as in dead battery condition, PCA9430 starts charging after 200 ms with default settings.

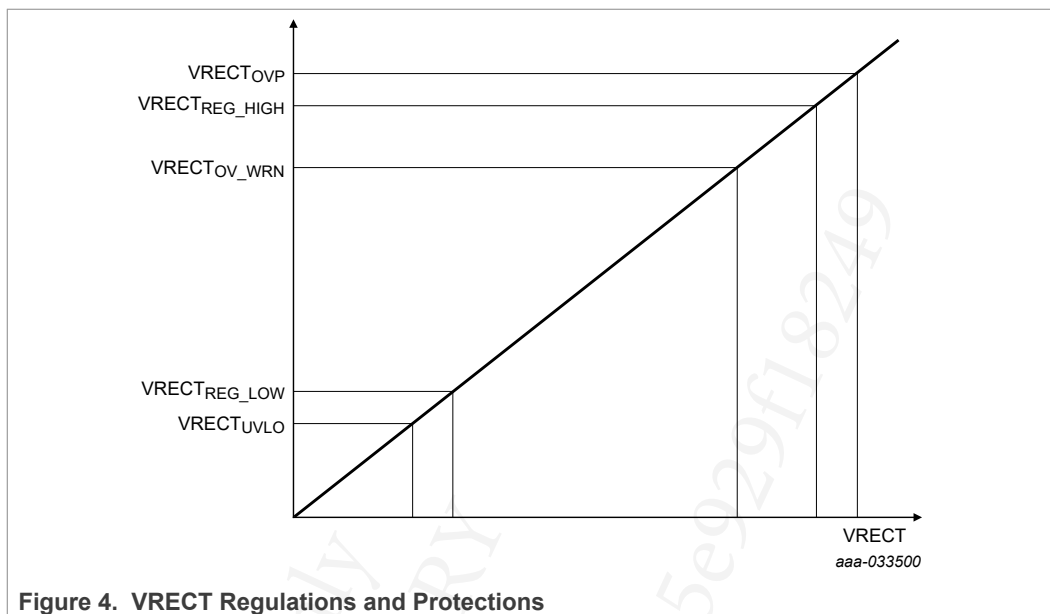
8.4 Integrated rectifier

PCA9430 uses an integrated rectifier to convert the AC voltage from the coil to a DC voltage on the RECT pins.

If higher efficiency is needed in the system, an external rectifier output can be connected to RECT1, AC1, and AC2. PCA9430 supports up to 200 mW charging with internal rectifier and up to 1 W charging with external rectifier.

8.5 VRECT voltage regulations and protections

The rectifier output voltage is regulated at a stable output for the following linear battery charger and LDO by the entire wireless charging control system. Furthermore, overvoltage protection circuit and undervoltage-lockout circuit are built in to protect VRECT when exceeding OVP threshold and UVLO threshold.



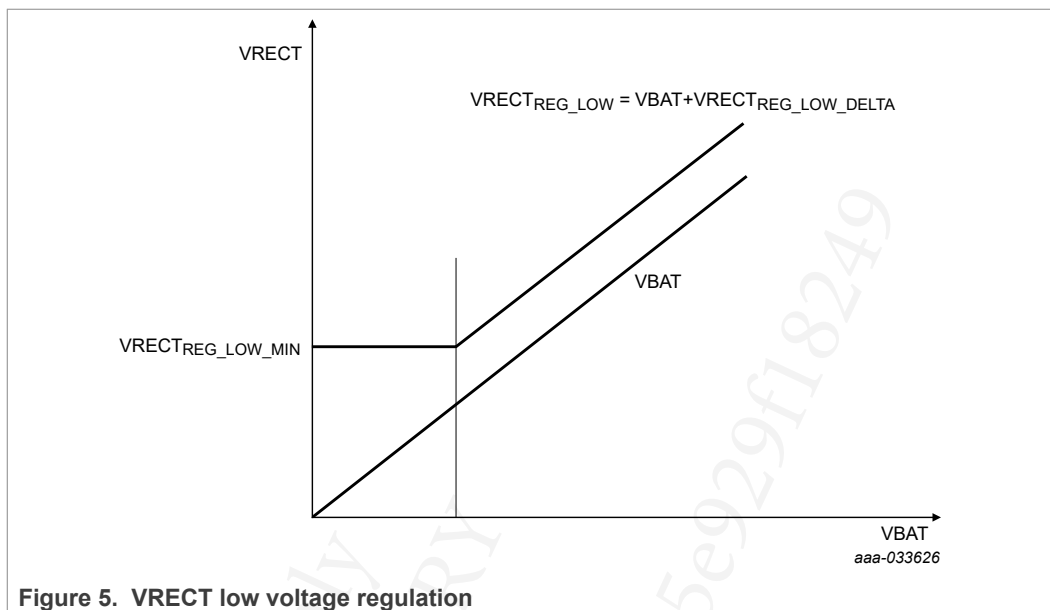
8.5.1 VRECT UVLO

The undervoltage-lockout circuit is integrated to properly shut down internal circuits when VRECT voltage drops below UVLO threshold. The threshold can be programmed by I²C register. When UVLO is triggered, an interrupt is triggered to notify the system.

8.5.2 VRECT low voltage regulation

When the required supply current from system is higher than the received current from antenna coils, the VRECT voltage decreases. A regulation circuit kicks off to maintain VRECT voltage above VRECT_{REG_LOW} threshold. The feature ensures the battery charging process not terminated because VRECT drops too low. In this case, an interrupt is issued to notify the system, so that it can negotiate higher power with transmitter.

The VRECT_{REG_LOW} threshold is defined versus the V_{BAT} voltage as shown in [Figure 5](#).

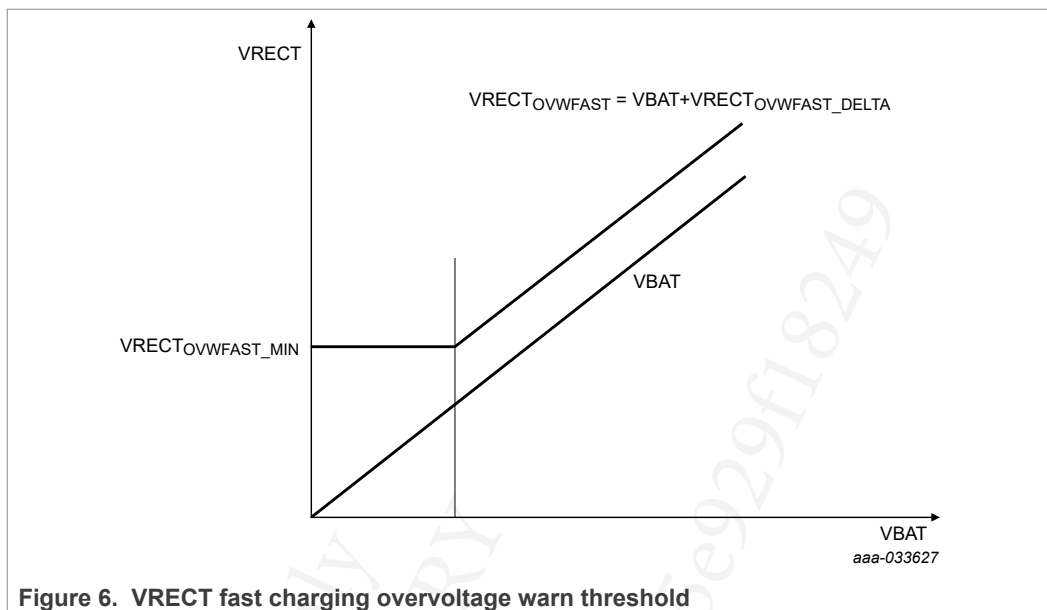


8.5.3 VRECT overvoltage warning threshold

The VRECT overvoltage warning threshold is set to notify the wireless charging system that the received power is larger than required power and VRECT voltage is ramping up. With this pre-warning interrupt, the system can actively negotiate with transmitter to reduce power and stabilize the VRECT voltage for higher wireless charging efficiency.

The VRECT warning voltage is set at the following two different conditions:

- When the linear charger is in precharge stages (dead battery charging, low battery charging and top off charging stage), the warning threshold is set by VRECT_OVWPRE.
- When the linear charger is in fast charging stage (CC and CV charging), the warning threshold is set by VRECT_OVWFAST_DELTA, starting from VRECT_OVWFAST_MIN. [Figure 6](#) shows the VRECT_OVWFAST_DELTA and VRECT_OVWFAST_MIN.



8.5.4 VRECT high voltage regulation

If VRECT keeps ramping up, the VRECT high voltage regulation circuit is triggered when VRECT exceeds $VRECT_{REG_HIGH}$. The circuit regulates the VRECT voltage at $VRECT_{REG_HIGH}$, and an interrupt is issued to notify the system.

8.5.5 VRECT overvoltage protection

The overvoltage protection circuit is integrated to protect against a sudden jump of voltage or transmitted power, such as WLC_L moving to better position.

The AC clamping method is equipped in PCA9430 for overvoltage protection. The circuit shorts AC1 and AC2 inputs directly to PGND when the RECT pins voltage exceeds overvoltage threshold.

During the overvoltage protection event, linear charger is disabled but internal digital and analog circuits are on.

8.6 Temperature sensing by NTC pin

With the temperature sensing pin, the external thermistor (NTC) is connected between NTC pin and ground. The thermistor may be included in the battery pack to monitor the battery pack temperature, or it may be an additional component user chooses to have on the board to monitor the temperature at a chosen area.

The voltage at NTC pin is monitored, and user can enable this feature through I²C-bus interface (NTC_EN) to implement JEITA compliant charging at a safe temperature. Per JEITA standard, there are four temperature thresholds:

- Cold threshold (T1, 0 °C as example)
- Cool threshold (T2, 10 °C as example)
- Warm threshold (T3, 45 °C as example)
- Hot threshold (T4, 60 °C as example)

Each of the above temperature threshold represents a voltage threshold. When the monitored temperature, T , falls into a different temperature zone, the charger should adjust the charging method accordingly:

- $T > T_4$ or $T < T_1$, i.e., when the temperature is in "cold" or "hot" zone, charging is suspended, so is the safety timer;
- $T_1 < T < T_2$, charging current is reduced by 50 % of the programmed current level;
- $T_2 < T < T_3$, normal charging;
- $T_3 < T < T_4$, the CV mode regulating voltage should be set as $V_{BAT_REG}[5:0] - \Delta V_{BAT_REG(HOT)}$

To disable this function, set NTC_EN to "0".

An accurate current source is supplied from the NTC pin. With a NTC thermistor connected between NTC and AGND pins, a voltage is generated and monitored by on-chip 10-bit ADC.

8.7 VPWR LDO output for dead-battery

In dead battery mode, the WLC_L charging receiver requires the host AP or MCU to wake up first to setup NFC link and WLC wireless charging protocol. PCA9430 provides a LDO output by VPWR pin to supply host AP or MCU at dead battery conditions. The maximum supply current is up to 50 mA, and the voltage can be adjusted by I²C register according to [Section 8.13.3.15](#).

When the WLC_L system wakes up from dead battery mode, the MCU power supply should be switched to system LDO from PMIC or dedicated LDO connected to battery.

The short circuit protection is integrated to protect the device. After short circuit protection is triggered, PCA9430 retries every 1 s.

In order to have system level flexibility, reverse current protection (RCP) circuit is integrated. This allows VPWR output to directly connect to an external LDO, with reverse current protection feature, to supply system MCU or AP.

8.8 Linear battery charger

The battery charger is a linear charger. The charging is done through a linear switch with the following output protections:

- Reverse current protection: trigger when $V_{RECT} < V_{BAT} + V_{RECT2BAT_HEADROOM}$. ($V_{RECT2BAT_HEADROOM} = 100$ mV, typical)
- Charging current limiting: a function of programmed threshold and battery temperature
- VBAT short circuit protection: when the linear charger enters dead battery charging at the beginning of charging cycle, a dead battery timer starts the count. If charging cycle enters low voltage charging stage before the dead battery timer expiration. The battery is not short and continue the charging. If the timer expired in dead battery stage, the battery is short and charging is stop. While the linear charger is charging the battery, VOUT has been shorten to ground suddenly for an abnormal reason. The analog short protection circuit kicks off and protects both the device and system. PCA9430 retries charging after 1 s and enters a new charging cycle. Battery short timer may expire if the short is not removed.

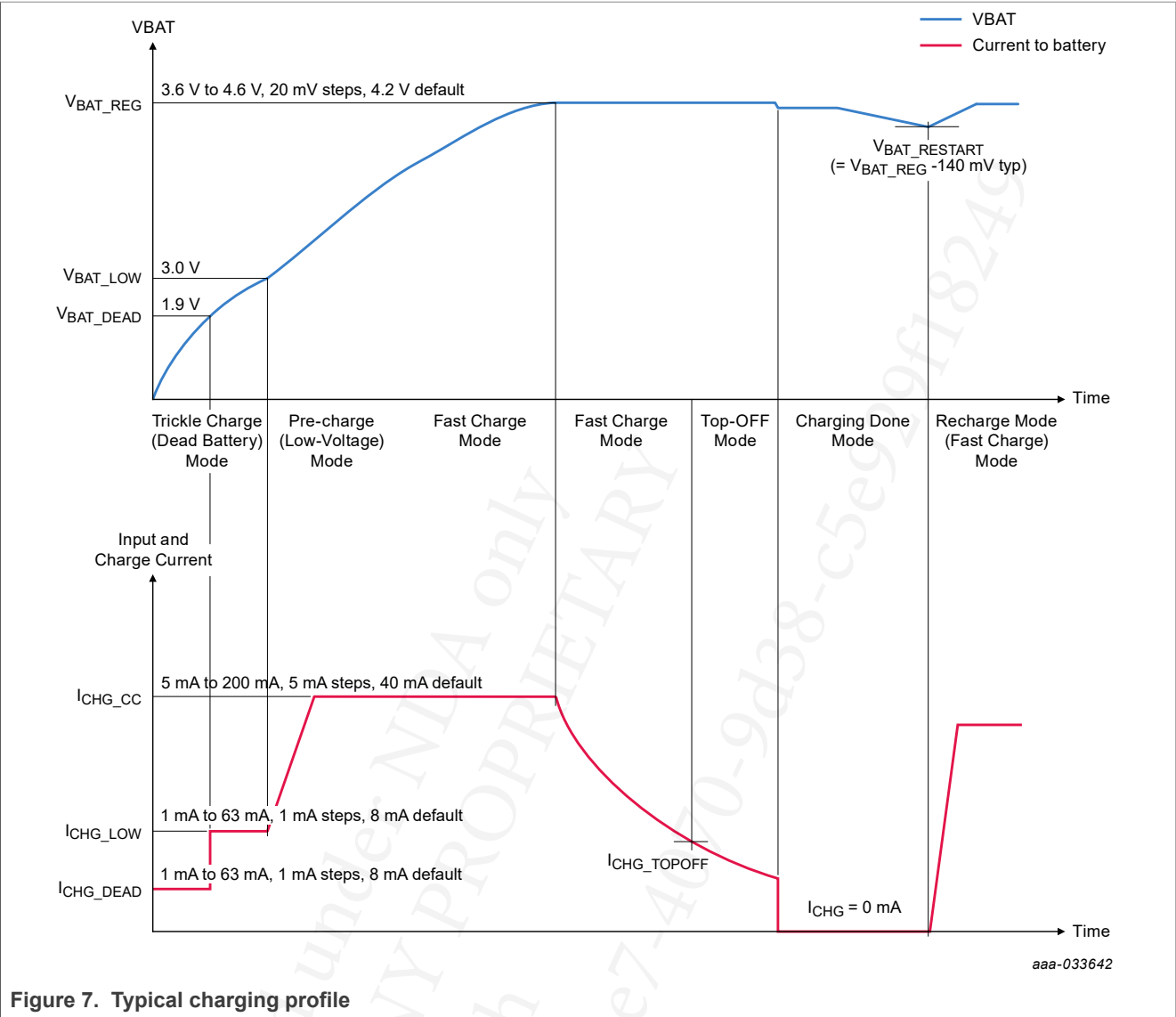
If the battery voltage is below the V_{BAT_LOW} threshold, the battery is considered discharged and a preconditioning cycle begins. The amount of pre-charge current can be programmed through I²C register setting as I_{CHG_DEAD} and I_{CHG_LOW} according to

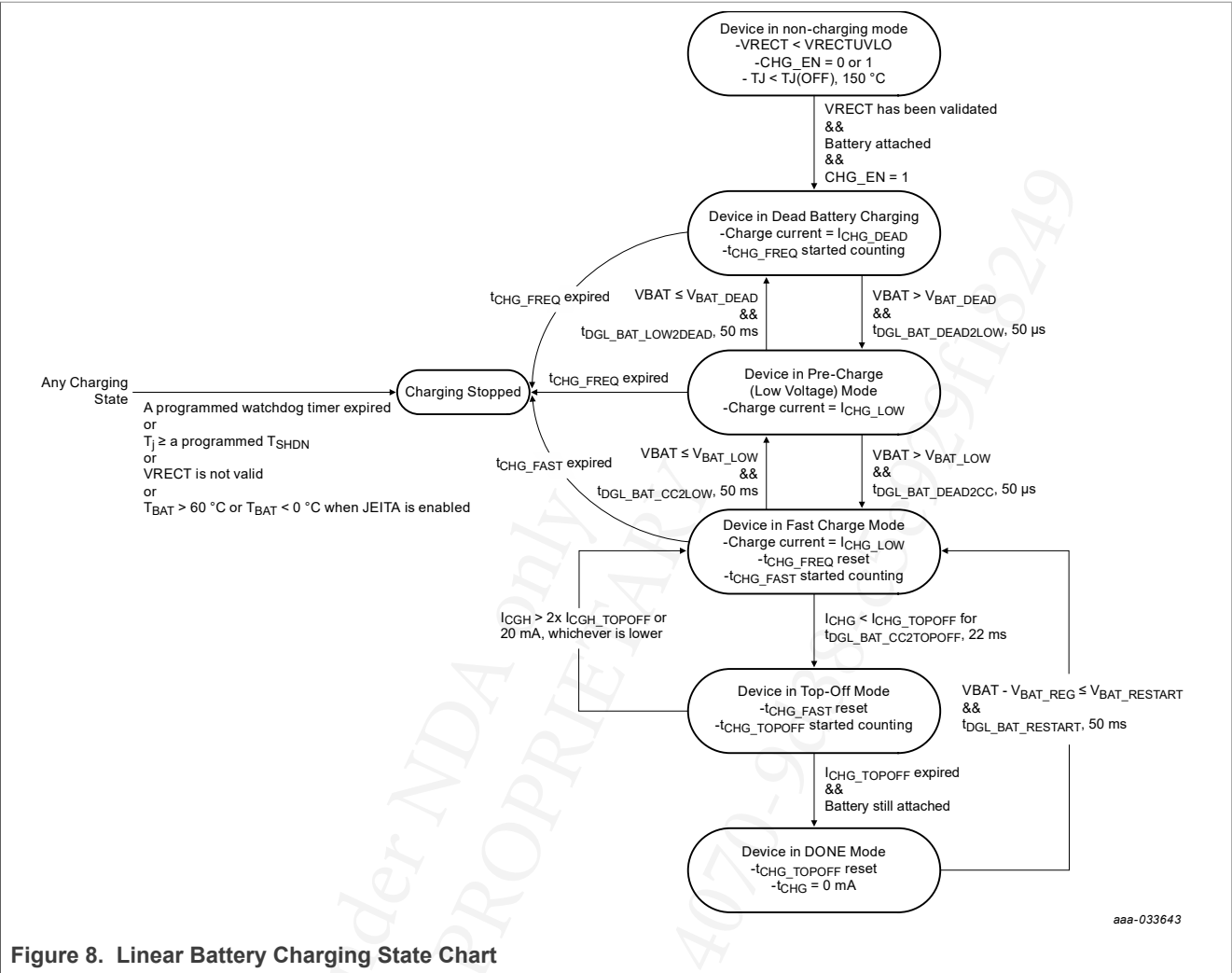
V_{BAT} voltage. This feature is useful when there is a load connected directly across the battery (at V_{OUT} pin) "stealing" the battery current. The pre-charge current can be set higher to account for the system loading while allowing the battery to be properly conditioned. Once the battery voltage has charged to the V_{BAT_LOW} threshold, fast charge is initiated and the fast charge current is applied. The fast charge constant current is programmed using I²C register. The constant current provides the bulk of the charge. Power dissipation in the IC is greatest in fast charge with a lower battery voltage.

If the IC reaches the programmed thermal regulation threshold temperature, the IC enters thermal regulation; it increases the safe-charging-timer period by 2x and reduces the charge current by half to keep the temperature from rising any further when battery charger works in constant current charging mode, or at a reduced regulated voltage when battery charger works in constant voltage charging mode. [Figure 7](#) shows the charging profile with thermal regulation. Once the cell has charged to the regulation voltage the voltage loop takes control and holds the battery at the regulation voltage until the current tapers to the top-off threshold.

8.8.1 Battery Charging Management

Battery charging management supports typical constant current/constant voltage charging profile for single cell Li-Ion battery, and also supports pre-qualification (dead battery, low battery), top-off mode, etc. Compliant with JEITA and thermal regulation.





8.8.2 Battery Temperature Sensing and JEITA-compliant Charging Profile

Operation Over NTC Bias Voltage (JEITA compliant).

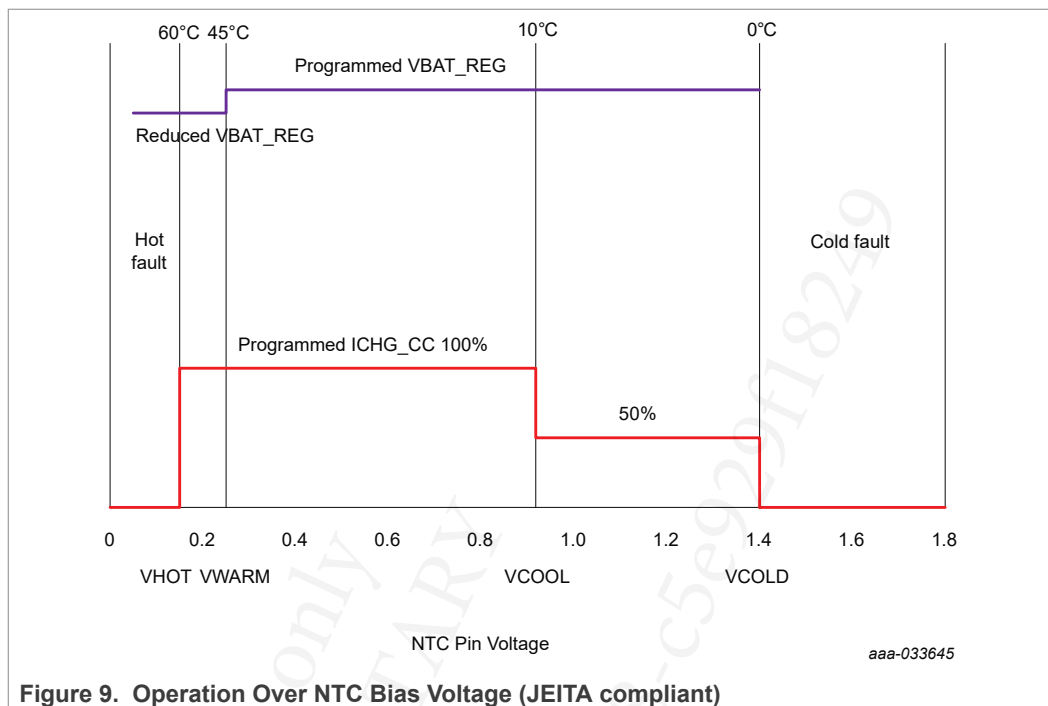


Figure 9. Operation Over NTC Bias Voltage (JEITA compliant)

8.8.3 Low-Battery/Dead-Battery (Pre-Qualification) Charging

If the battery is detected and $V_{BAT} < V_{BAT_LOW}$, the charger initiates the pre-charge sequence using a predefined (I^2C register) current.

When under dead-battery condition, the charging current (I_{CHG_DEAD}) is programmed by $ICHG_DEAD[5:0]$; when under low-battery, the charging current (I_{CHG_LOW}) is programmed by $ICHG_LOW[5:0]$. Once $V_{BAT} \geq V_{BAT_LOW}$, the charger moves to the next state.

8.8.4 Constant Current Charging/Constant Voltage Charging (Fast Charging) and Termination

When $V_{BAT} \geq V_{BAT_LOW}$, the charger enters Constant Current (CC) charging state. In this state, the battery voltage V_{BAT} continues to rise, while the battery is being charged with the current set by $ICHG_CC[5:0]$, until V_{BAT} reaches the maximum allowable voltage set by $VBAT_REG[5:0]$.

At this time, the charger enters the Constant Voltage (CV) mode. While operating in the CV mode, the voltage is still regulated at the level set by $VBAT_REG[5:0]$, and the charging current continues to decrease.

When the charging current drops below the top-off current threshold, set by $ICHG_TOPOFF[5:0]$, the charger enters TOPOFF mode; upon expiration of TOPOFF timer (set by $T_TOPOFF[1:0]$), the charger enters DONE mode.

8.8.5 Charging Safety Timer

Two sets of charging safety timers are implemented on PCA9430UK. These timers ensure the charging is terminated if the charging time is longer than its predefined limit at given states:

- Pre-qualification timer, set by ICHG_PREQ_TIMER[1:0]: 15 min ~ 60 min
- Fast charge timer, set by ICHG_FAST_TIMER[1:0]: 3 hr ~ 9 hr

When the charging safety timer expires, an interrupt is filed to notify the system; the charger turns off and CHG_EN is set to "0" automatically. To restart charging the system needs to read and clear the safety timer expiration interrupt and CHG_EN needs to be "1".

8.8.6 Refresh charging

While in DONE mode, if $V_{BAT} < V_{BAT_REG} - V_{BAT_RESTART}$, the battery charger resumes Constant Current (CC) Mode.

8.8.7 Start New Charge Cycle

When VRECT exceeds UVLO threshold, battery attached to VOUT, or CHG_EN set to "1", the device initializes a new charging process.

8.8.8 Battery Attach Detection

Battery attach detection is integrated in PCA9430. The detection flow is shown in [Figure 10](#). When the battery is detected as "Present", the battery charger goes back to normal charging flow. When the battery is detected as "Absent", the battery charger continuously runs the detection flow with a delay in between.

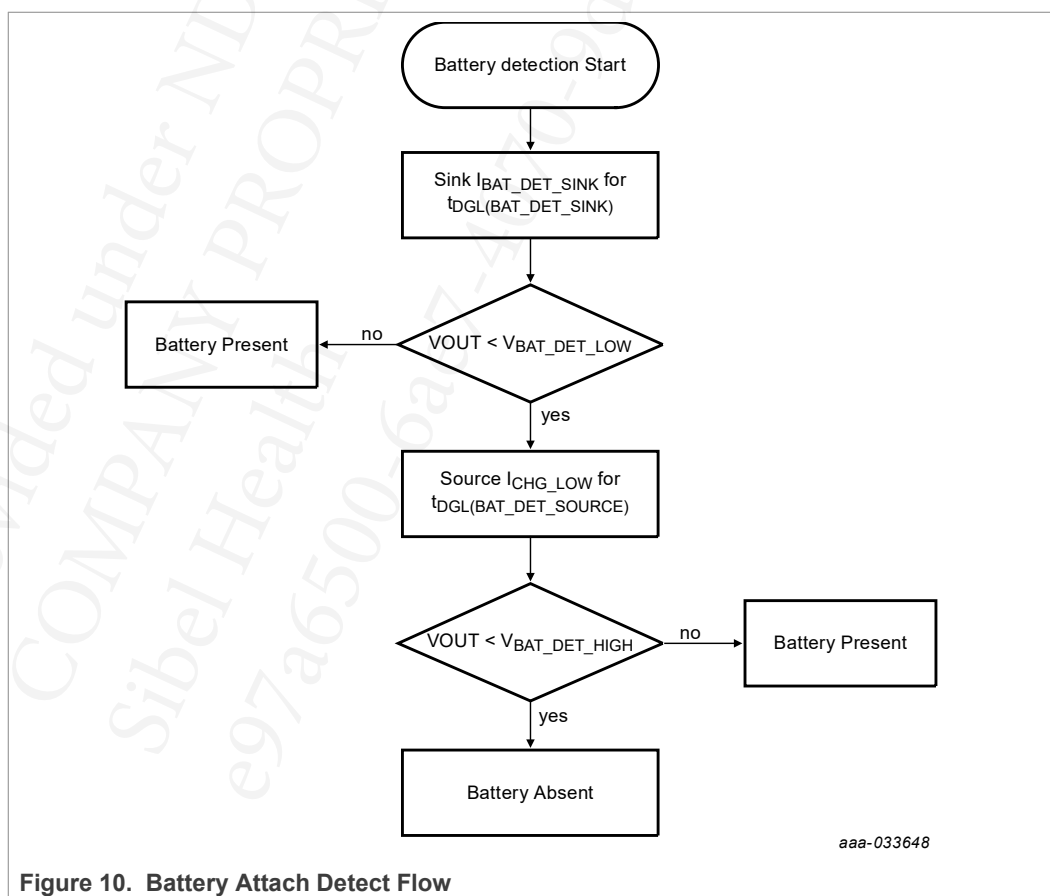


Figure 10. Battery Attach Detect Flow

8.9 RXIR function

An RXIR function is integrated in PCA9430, providing an interrupt function for the wireless charger receiver side to wireless charger transmitter side. When the receiver side AP or MCU wants to notify the wireless charger transmitter side, the AP or MCU can configure the RXIR registers and trigger the RXIR. When the wireless charger transmitter side receives the RXIR request, the transmitter reads the NDEF in CRN120 chip of receiver side and responds to the request.

8.10 ADC

The 10-bit ADC is integrated to measure the parameters for system control needs, enabling direct measurements of internal voltages, currents, on-chip temperatures, and off-chip temperatures (VRECT, IRECT, VOUT, IOUT, chip temperature and NTC voltage). The measured results are stored in the I²C register for host MCU or AP access.

The integrated multi-channel ADC can operate in three different modes: single-shot mode, continuous conversion mode, and burst mode. All three modes support an 8-sample averaging. The measured channel can be selected in the SAMPLES_EN register.

If averaging is enabled, the accumulator is cleared first, before the ADC samples a channel 8 times to collect fresh data. The average value is calculated from this set of values and stored in a 10-bit register which is used for ADC I²C register value as described before.

In one shot mode, once the ADC value is available, the ADC_EN bit is cleared.

8.11 Varactor tuning DAC

The resonator tuning function is supported by an integrated 5-bit varactor tuning DAC. It receives a 5-bit control code from host MCU or AP and generates a continuous VTUNE voltage to control external varactor.

8.12 On-chip over temperature protection

When PCA9430 is conducting and the device temperature exceeds the set temperature threshold, the on-chip over-temperature protection (OTP) circuit disables the rectifier, linear battery charger and VPWR LDO. An interrupt is sent to the host for notification. The device will not be re-enabled in over-temperature protection. Once the device temperature decreases to 20 °C below the threshold, the device returns to normal operation. The temperature shutdown threshold can be set by [Section 8.13.3.10](#).

8.13 I²C-bus interface and register

The PCA9430UK implements an I²C-bus slave interface to communicate with the host system. The interface supports Fast Mode plus with up to 1 Mbit/s. A detailed description of the I²C-bus specification is given in UM10204, Rev. 06, 4 April 2014, "I²C-bus specification and user manual".

Features such as clock-stretching, 10-bit slave address and general call are not supported. Auto increment with address wrap-around is supported.

8.13.1 I²C Device Address

Following a START condition, the bus master must send the target slave address followed by a read (R/W = 1) or write (R/W = 0) operation bit. The slave address of the PCA9430 is shown in [Table 5](#).

Table 5. PCA9430 address

A7	A6	A5	A4	A3	A2	A1	A0
1	1	1	0	0	0	0	0/1
Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed	R/W

8.13.2 Command register and address auto-increment

Following the successful acknowledgment of the slave address and a write bit, the bus master sends a byte to the PCA9430. This byte is stored in the Command register.

At power-up, the Command register defaults to 00h; the master can set the AI bit to '1' and the lowest seven bits to '0' for auto-increment address. The lowest six bits are used as a pointer to determine which register is accessed. Only a command register code with the address from 0x00 to 0x3C are acknowledged. The registers are defined in [Table 6](#) and some of the registers are undefined for user but reserved for internal use.

The most significant bit of the Command register is for Auto-Increment. If the Auto-Increment flag is set, the six low-order bits of the Control register are automatically incremented after a read or write. This allows the user to program the registers sequentially. The contents of these bits roll over to '00 0000' after the last register (address = 0x3Ch) is accessed. Only the six least significant bits are affected by the AI flag. Unused bits must be programmed with zeros.

Example:

Reading all ADC values is initiated by sending the device address, followed by the VRECT_ADC_H byte address (30h or '011 0000'b) increased by 00h or '0000 0000'b to B0h or '1011 0000'b. After this altered address is sent, the continuous reading sequence can be executed without resending the device or register address until the master sends a stop condition. As the device supports address wrap-around, the address pointer is set to 00h after a read operation is executed on the last available register address 0x3Ch.

For these ADC value related read-only registers with address range from 0x30 to 0x3C (except for 0x3B register as shown in [Table 6](#)), when performing the read command for a certain ADC channel, it always should start from the XXX_ADC_H register first, then perform read command to XXX_ADC_L register. For example, when reading the ADC value for VRECT, start the read command to read register VRECT_ADC_H (address 0x30) first and then read from VBUS_ADC_L (address 0x31). For other registers with address from 0x30~0x3C, the same read sequence should be applied.

8.13.3 I²C Register Map

The registers of PCA9430 are listed in [Table 6](#).

Table 6. PCA9430 register table

Address (Hex)	Register name	Description	Access	Reset Value (Binary)
System control registers				

Table 6. PCA9430 register table...continued

Address (Hex)	Register name	Description	Access	Reset Value (Binary)
00h	Device ID	Device information and revision	R	00001000
01h	SYSTEM_INT	Interrupt for system level	R/C	00000000
02h	SYSTEM_INT_MASK	System interrupt mask register	R/W	00000000
03h	VRECT_INT	Interrupt for VRECT protections and regulations	R/C	00000000
04h	VRECT_INT_MASK	VRECT interrupt mask register	R/W	00000000
05h	BATCHG_INT	Interrupt for battery charger	R/C	00000000
06h	BATCHG_INT_MASK	Battery charger Interrupt Mask mask register	R/W	00000000
08h	VRECT_THD	VRECT voltage threshold and regulation setting register	R/W	00000000
09h	VRECT_OVW	VRECT overvoltage warning setting register	R/W	00010000
0Ah	TEMP_THD	Temperature threshold setting register	R/W	00000000
0Bh	WD_EN_RST	Watchdog, soft reset and enable signal register	R/W	00000010
0Ch	Varactor_DAC	Varactor DAC control register	R/W	00000000
0Dh	ADC_CONTROL	ADC control register	R/W	00000110
0Eh	Sample_EN	ADC channel select register	R/W	01111111
0Fh	VPWR_CONFIG	Configure VPWR LDO	R/W	10000011
10h	RXIR_CONFIG	Configure RXIR function	R/W	00000000
Battery charger control registers				
20h	BQ_CONTROL	Battery charger control register	R/W	10101010
21h	CC_CURRENT	CC charging current setting, CHG_LOCK register	R/W	00001000
22h	TP_CURRENT	Top off charging current setting, CHG_LOCK register	R/W	00000100
23h	LB_CURRENT	Low battery charging current setting, CHG_LOCK register	R/W	00001000
24h	DB_CURRENT	Dead battery charging current setting, CHG_LOCK register	R/W	01000100
25h	VBAT_REG	Regulated battery voltage and restart battery voltage, CHG_LOCK register	R/W	00011110
26h	BQTIMER_AND_NTC	Battery charger timer setting and NTC thermistor resistance setting, CHG_LOCK register	R/W	10010101
27h	BQTHM_AND_SHRT	Battery charger thermal and VBAT short circuit setting register	R/W	00100100
28h	Battery charger status 1	Battery charger status register 1	R/W	00000000

Table 6. PCA9430 register table...continued

Address (Hex)	Register name	Description	Access	Reset Value (Binary)
29h	Battery charger status 2	Battery charger status register 2	R/W	00000000
2Ah	Battery charger status 3	Battery charger status register 3	R/W	00000000
2Bh	Battery charger status 4	Battery charger status register 4	R/W	00000000
ADC registers				
30h	VRECT_ADC_H	Rect 10-bit ADC high byte	R	00000000
31h	VRECT_ADC_L	Rect 10-bit ADC low byte	R	00000000
32h	VTUNE_ADC_H	VTUNE 10-bit ADC high byte	R	00000000
33h	VTUNE_ADC_L	VTUNE 10-bit ADC low byte	R	00000000
34h	VOUT_ADC_H	VOUT 10-bit ADC high byte	R	00000000
35h	VOUT_ADC_L	VOUT 10-bit ADC low byte	R	00000000
36h	IOUT_ADC_H	IOUT 10-bit ADC high byte	R	00000000
37h	IOUT_ADC_L	IOUT 10-bit ADC low byte	R	00000000
38h	IRECT_ADC_H	IRECT 10-bit ADC high byte	R	00000000
39h	IRECT_ADC_L	IRECT 10-bit ADC low byte	R	00000000
3Ah	TDIE_ADC_H	On chip temperature 10-bit ADC high byte	R	00000000
3Bh	NTC_ADC_H	NTC voltage 10-bit ADC high byte	R	00000000
3Ch	NTC_ADC_L	NTC voltage 10-bit ADC low byte	R	00000000

8.13.3.1 Device ID Register (Address 00h)

The device identification code stores a unique identifier for each version and/or revision of a PCA9430, so that the connected MCU recognizes it automatically.

This is a READ ONLY register.

Table 7. Device ID Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R	Reserved
6	ID [3]	0	R	Device ID
5	ID [2]	0	R	
4	ID [1]	0	R	
3	ID [0]	1	R	
2	Revision[2]	0	R	Revision number
1	Revision[1]	0	R	
0	Revision[0]	1	R	

8.13.3.2 SYSTEM_INT Register (Address 01h)

The System interrupt register contains flags indicating an overvoltage and over temperature event. An event is latched and only the first occurrence triggers an interrupt

(if not masked). Reoccurring events do not change the flags' status or trigger an additional interrupt. The interrupt registers are automatically cleared by reading if not masked through their masking bits in [Section 8.13.3.3](#).

To check the actual status of an interrupt, the interrupt registers have to read twice, as the event setting a flag might no longer be valid.

This is a READ ON CLEAR register.

Table 8. SYSTEM_INT Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/C	Reserved
6	RSVD	0	R/C	Reserved
5	RSVD	0	R/C	Reserved
4	RSVD	0	R/C	Reserved
3	VPWR_SCP_INT	0	R/C	VPWR short circuit protection Interrupt: 0: The VPWR short circuit protection is not triggered 1: The VPWR short circuit protection triggered
2	WD_TIMER_INT	0	R/C	Watchdog Timer Expiration Interrupt: 0: The watchdog timer expiration has not happened since the last time this bit was cleared. 1: The watchdog timer expiration has happened since the last time this bit was cleared.
1	Temp_WRN_INT	0	R/C	Die temp pre-warning interrupt: 1: die temp $\geq$ TWRN; 0: die temp $<$ TWRN TWRN threshold is configured by TEMP_WRN[1:0]
0	Temp_SHD_INT	0	R/C	Thermal shutdown interrupt: 0: thermal shutdown is not triggered 1: die temp $\geq$ TSTDN (set in THM_SHDN[2:0], thermal shutdown is triggered

8.13.3.3 SYSTEM_INT_MASK Register (Address 02h)

The SYSTEM_INT_MASK register enables the masking (disabling) of the different available interrupt signals of register SYSTEM_INT. When enabled, interrupt events trigger the INTB pin to be pulled low when matching the flag bit in the interrupt register SYSTEM_INT. The interrupt registers are automatically cleared by reading them. The INTB pin has to be connected through an external pull-up resistor to VDD of the controlling MCU.

The register can be reset to default by writing a '1' to the REG_RST bit in the CONTROL register.

This is a READ/WRITE register.

Table 9. SYSTEM_INT_MASK Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	RSVD	0	R/W	Reserved
5	RSVD	0	R/W	Reserved

Table 9. SYSTEM_INT_MASK Register...continued

Bit	Symbol	Default Value	Type	Function
4	RSVD	0	R/W	Reserved
3	VPWR_SCP_INT_MSK	0	R/W	VRECT voltage low interrupt disable: 0: VPWR_SCP_INT can trigger INT 1: VPWR_SCP_INT trigger disabled
2	WD_TIMER_INT_MSK	0	R/W	Watchdog timer interrupt disable: 0: WD_TIMER_INT can trigger INT 1: WD_TIMER_INT trigger disabled
1	Temp_WRN_INT_MSK	0	R/W	Temperature warning interrupt disable: 0: Temp_WRN_INT can trigger INT 1: Temp_WRN_INT trigger disabled
0	Temp_SHD_INT_MSK	0	R/W	Temperature shut down interrupt disable: 0: Temp_SHD_INT can trigger INT 1: Temp_SHD_INT trigger disabled

8.13.3.4 VRECT_INT Register (Address 03h)

The VRECT interrupt register contains flags indicating the protections and regulations alert of VRECT control. An event is latched and only the first occurrence triggers an interrupt (if not masked). Reoccurring events will not change the flag's status or trigger an additional interrupt. The interrupt registers are automatically cleared by reading if not masked through their masking bits in [Section 8.13.3.5](#).

To check the actual status of an interrupt, the interrupt registers have to read twice, as the event setting a flag might no longer be valid.

This is a READ ON CLEAR register.

Table 10. VRECT_INT Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/C	Reserved
6	VRECT_RDY_INT	0	R/C	VRECT voltage ready Interrupt: 0: The VRECT voltage is less than V_{UVLO} 1: The VRECT voltage is larger than V_{UVLO}
5	VRECT_GOOD_INT	0	R/C	VRECT voltage good Interrupt: 0: The VRECT voltage is not in the range of $V_{UVLO} < VRECT < V_{OVP}$ 1: The VRECT voltage is in the range of $V_{UVLO} < VRECT < V_{OVP}$
4	VRECT_REGHIGH_INT	0	R/C	VRECT high voltage regulation trigger interrupt: 0: VRECT high voltage regulation not triggered 1: VRECT high voltage regulation triggered
3	VRECT_REGLOW_INT	0	R/C	VRECT low voltage regulation trigger interrupt: 0: VRECT low voltage regulation not triggered 1: VRECT low voltage regulation triggered

Table 10. VRECT_INT Register...continued

Bit	Symbol	Default Value	Type	Function
2	VRECT_OVWFAST_INT	0	R/C	VRECT overvoltage warning interrupt: 0: VRECT fast charging overvoltage warning not triggered 1: VRECT fast charging overvoltage warning triggered
1	VRECT_OVWPRE_INT	0	R/C	VRECT pre-charging overvoltage warning interrupt: 0: $VRECT < VRECT_OVWPRE$ 1: $VRECT \geq VRECT_OVWPRE$ VRECTWRN threshold is configured by RECT_OV_WRN[2:0]
0	VRECT_OVP_INT	0	R/C	VRECT overvoltage protection interrupt: 0: VRECT overvoltage protection not triggered 1: $VRECT > OVPRECT$ (set in RECT_OVP[2:0]), overvoltage protection triggered

8.13.3.5 VRECT_INT_MASK Register (Address 04h)

The VRECT_INT_MASK register enables the masking (disabling) of the different available interrupt signals of register VRECT_INT. When enabled, interrupt events trigger the INTB pin to be pulled low when the matching flag bit in the interrupt register VRECT_INT is set. The interrupt registers are automatically cleared by reading them. The INTB pin has to be connected through an external pull-up resistor to VDD of the controlling MCU.

The register can be reset to default by writing a '1' to the REG_RST bit in the CONTROL register.

This is a READ/WRITE register.

Table 11. VRECT_INT_MASK Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
5	VRECT_RDY_INT_MSK	0	R/W	VRECT voltage ready interrupt disable: 0: VRECT_RDY_INT can trigger INT 1: VRECT_RDY_INT trigger disabled
4	VRECT_GOOD_INT_MSK	0	R/W	VRECT voltage good interrupt disable: 0: VRECT_GOOD_INT can trigger INT 1: VRECT_GOOD_INT trigger disabled
3	VRECT_REGHIGH_INT_MSK	0	R/W	VRECT high voltage regulation trigger interrupt disable: 0: VRECT_REGHIGH_INT can trigger INT 1: VRECT_REGHIGH_INT trigger disabled
2	VRECT_REGLOW_INT_MSK	0	R/W	VRECT low voltage regulation trigger interrupt disable: 0: VRECT_REGLOW_INT can trigger INT 1: VRECT_REGLOW_INT trigger disabled
1	VRECT_OVWFAST_INT_MSK	0	R/W	Fast charging overvoltage warning interrupt disable: 0: VRECT_OVWRN_INT can trigger INT 1: VRECT_OVWRN_INT trigger disabled

Table 11. VRECT_INT_MASK Register...continued

Bit	Symbol	Default Value	Type	Function
1	VRECT_OVWPRES_MSK	0	R/W	Pre-charging overvoltage warning interrupt disable: 0: VRECT_OVWRN_INT can trigger INT 1: VRECT_OVWRN_INT trigger disabled
0	VRECT_OVP_INT_MSK	0	R/W	Overvoltage protection interrupt disable: 0: VRECT_OVP_INT can trigger INT 1: VRECT_OVP_INT trigger disabled

8.13.3.6 BATCHG_INT Register (Address 05h)

The battery charger interrupt register contains flags indicating an overvoltage and over temperature event. An event is latched and only the first occurrence triggers an interrupt (if not masked). Reoccurring events will not change the flags' status or trigger an additional interrupt. The interrupt registers are automatically cleared by reading if not masked through their masking bits in [Section 8.13.3.7](#).

To check the actual status of an interrupt, the interrupt registers have to be read twice, as the event setting a flag might no longer be valid.

This is a READ ON CLEAR register.

Table 12. BATCHG_INT Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/C	Reserved
6	VOUT_SHORT_INT	0	R/C	Analog short circuit protection Interrupt: 0: Analog short protection circuit does not detect VOUT short 1: Analog short protection circuit detects VOUT short
5	ICRG_DEAD_TIMER_INT	0	R/C	Battery short detection timer Expiration Interrupt: 0: The battery short detection timer expiration has not happened since the last time this bit was cleared. Battery is not short 1: The battery short detection timer expiration has happened since the last time this bit was cleared. Battery is short
4	ICRG_FAST_TIMER_INT	0	R/C	Fast Charging Timer Expiration Interrupt: 0: The fast charging timer expiration has not happened since the last time this bit was cleared. 1: The fast charging timer expiration has happened since the last time this bit was cleared.
3	ICRG_PREQ_TIMER_INT	0	R/C	Pre-qualification Charging Timer Expiration Interrupt: 0: The pre-qual charging timer expiration has not happened since the last time this bit was cleared. 1: The pre-qual charging timer expiration has happened since the last time this bit was cleared.
2	VBAT_DET_INT	0	R/C	Battery presence Interrupt: 0: The VBAT_DET_OK bit has not changed since the last time this bit was cleared. 1: The VBAT_DET_OK bit has changed since the last time this bit was cleared.

Table 12. BATCHG_INT Register...continued

Bit	Symbol	Default Value	Type	Function
1	VBAT_INT	0	R/C	Battery Interrupt: 0: The VBAT_OK bit has not changed since the last time this bit was cleared. 1: The VBAT_OK bit has changed since the last time this bit was cleared.
0	CHG_OK_INT	0	R/C	Charger Status Interrupt: 0: The CHG_OK bit has not changed since the last time this bit was cleared. 1: The CHG_OK bit has changed since the last time this bit was cleared

8.13.3.7 BATCHG_INT_MASK Register (Address 06h)

The battery charger interrupt mask register enables the masking (disabling) of the different available interrupt signals of register BATCHG_INT. When enabled, interrupt events trigger the INTB pin to be pulled low when the matching flag bit in the event register BATCHG_INT is set. The interrupt registers are automatically cleared by reading them. The INTB pin has to be connected through an external pull-up resistor to VDD of the controlling MCU.

The register can be reset to default by writing a '1' to the REG_RST bit in the CONTROL register.

This is a READ/WRITE register.

Table 13. BATCHG_INT_MASK Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	VOUT_SHORT_INT_MASK	0	R/W	Analog short circuit protection Interrupt Mask: 0: VOUT_SHORT_INT is not masked 1: VOUT_SHORT_INT is masked
5	ICRG_DEAD_TIMER_INT_MASK	0	R/W	Battery short detection timer Expiration Interrupt Mask: 0: ICRG_DEAD_TIMER_INT is not masked 1: ICRG_DEAD_TIMER_INT is masked
4	ICRG_FAST_TIMER_INT_MASK	0	R/W	Fast Charging Timer Expiration Interrupt Mask: 0: ICRG_FAST_TIMER_INT is not masked 1: ICRG_FAST_TIMER_INT is masked
3	ICRG_PREQ_TIMER_INT_MASK	0	R/W	Pre-qual Charging Timer Expiration Interrupt Mask: 0: ICRG_PREQ_TIMER_INT is not masked 1: ICRG_PREQ_TIMER_INT is masked
2	VBAT_DET_INT_MASK	0	R/W	Battery presence Interrupt Mask: 0: VBAT_DET_INT is not masked 1: VBAT_DET_INT is masked
1	VBAT_INT_MASK	0	R/W	Battery Interrupt Mask: 0: VBAT_INT is not masked 1: VBAT_INT is masked

Table 13. BATCHG_INT_MASK Register...continued

Bit	Symbol	Default Value	Type	Function
0	CHG_OK_INT_MASK	0	R/W	Charger Interrupt Mask: 0: CHG_OK_INT is not masked 1: CHG_OK_INT is masked

8.13.3.8 VRECT_THD Register (Address 08h)

The VRECT_THD register provides VRECT voltage control for host AP or MCU.

This is a READ and WRITE register.

Table 14. RECT_THD Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	RSVD	0	R/W	Reserved
5	VRECT_UVLO[1:0]	00	R/W	Set RECTx pins voltage UVLO trip level: 00: 3.0 V; 01: 3.1 V; 10: 3.2 V; 11: 3.3 V
4				
3	RSVD	0	R/W	Reserved
2	RSVD	0	R/W	Reserved
1	RSVD	0	R/W	Reserved
0	RSVD	0	R/W	Reserved

8.13.3.9 VRECT_OVW Register (Address 09h)

The VRECT_OVW register provides VRECT overvoltage warning control for host AP or MCU.

This is a READ and WRITE register.

Table 15. RECT_THD Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	RSVD	0	R/W	Reserved
5	VRECT_OVWFAST_DELTA[1:0]	01	R/W	Set fast charging VRECT overvoltage warning delta to VBAT: 00: 0.5 V; 01: 0.75 V; 10: 1 V; 11: 1.25 V
4				
3	VRECT_OVWFAST_MIN[1:0]	00	R/W	Set fast charging VRECT overvoltage warning minimum level: 00: 4 V; 01: 4.2 V; 10: 4.4 V; 11: 4.6 V
2				
1	VRECT_OVWPRE[1:0]	00	R/W	Set pre-charge VRECT overvoltage warning level: 00: 7 V; 01: 7.25 V; 10: 7.5 V; 11: 7.75 V
0				

8.13.3.10 TEMP_THD Register (Address 0Ah)

The TEMP_THD register provides temperature threshold setting for host AP or MCU.

This is a READ and WRITE register.

Table 16. TEMP_THD Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	RSVD	0	R/W	Reserved
5	RSVD	0	R/W	Reserved
4	TEMP_WRN[1:0]	00	R/W	Die temperature warning threshold setting: 00: 80 °C 01: 90 °C 10: 100 °C 11: 110 °C
3				
2	TEMP_SHDN[2:0]	000	R/W	Thermal shutdown temperature threshold (rising, 20 °C hysteresis) setting: 000: 120 °C 010: 130 °C 100: 140 °C 110: 150 °C 001: 125 °C 011: 135 °C 101: 145 °C 111: 150 °C
1				
0				

8.13.3.11 WD_EN_RST Register (Address 0Bh)

The WD_EN_RST register provides watchdog control, soft reset and enable signal for host AP or MCU.

This is a READ and WRITE register.

Table 17. WD_EN_RST Register

Bit	Symbol	Default Value	Type	Function
7	WATCH_DOG[1]	00	R/W	Watchdog timer setting. Any valid R/W access acknowledged by the PCA9430, I ² C-bus clears the watchdog timer. When watchdog expires, it resets all I ² C register bits 00: watchdog timer disabled 01: 16 s 10: 32 s 11: 64 s
6	WATCH_DOG[0]			
5	RSVD	0	R/W	Reserved
4	ECO_EXIT	0	R/W	Exit ECO_Mode bit: 0: The device does not exit ECO_Mode 1: The device exits ECO_Mode
3	RSVD	0	R/W	Reserved
2	VARACTOR_EN	0	R/W	VARACTOR 5-bit control setting bit: 0: Varactor is off 1: Varactor is on
1	BQ_CHG_EN	1	R/W	Enable the linear battery charger: 0: linear battery charger is disabled 1: linear battery charger is enabled
0	SW_RST	0	W/C	I ² C register bit to its default setting, and meanwhile, this bit is cleared and reset back to "0" as well.

8.13.3.12 Varactor_DAC Register (Address 0Ch)

The Varactor_DAC register provides 5-bit control to external varactor through VTUNE.

This is a READ and WRITE register.

Table 18. Varactor Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	RSVD	0	R/W	Reserved
5	RSVD	0	R/W	Reserved
4	Varactor_DAC[4:0]	00000	R/W	VARACTOR DAC 5-bit setting, please refer to Section 8.11 "Varactor tuning DAC"
3				
2				
1				
0				

8.13.3.13 ADC_CONTROL Register (Address 0Dh)

The ADC_CTRL register controls the ADC operation modes and the die temperature sampling channel. The ADC offers continuous or single-shot sampling, 1-sample measurement, 8-sample averaging and 16-sample averaging.

The register can be reset to default by writing a '1' to the REG_RST bit in the WD_EN_RST register.

This is a READ/WRITE register.

Table 19. ADC_CONTROL Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	RSVD	0	R/W	Reserved
5	RSVD	0	R/W	Reserved
4	ADC_EN	0	R/W	ADC enable/disable 0: disable ADC 1: enable ADC ADC_EN is reset to 0 after conversion cycle 1-shot conversion mode was performed
3	ADC_RATE[1]	01	R/W	ADC sample mode control 00: 1-shot conversion 01: continuous conversion 1x: Burst mode, measure all opened channel every 1 s Channel control is defined in Section 8.13.3.14 "SAMPLE_EN Register (Address 0Eh)"
2	ADC_RATE[0]		R/W	
1	ADC_AVG_EN	1	R/W	ADC averaging enable/disable 0: disable averaging, one instantaneous measurement 1: enable averaging
0	RSVD	0	R/W	Reserved

8.13.3.14 SAMPLE_EN Register (Address 0Eh)

The SAMPLES_EN register controls the ADC channels selected for sampling, except the

die temperature sampling which is controlled in [Section 8.13.3.13](#). Whether a channel is selected or not does not influence any protection or regulation feature associated with that parameter.

The register can be reset to default by writing a '1' to the REG_RST bit in the WD_EN_RST register.

This is a READ/WRITE register.

Table 20. SAMPLE_EN Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	VTUNE_ADC_EN	1	R/W	0: disable VTUNE sampling 1: enable VTUNE sampling
5	VRECT_ADC_EN	1	R/W	0: disable VRECT sampling 1: enable VRECT sampling Selection does not influence VRECT protections
4	IRECT_ADC_EN	1	R/W	0: disable IRECT sampling 1: enable IRECT sampling Selection does not influence IRECT protections and/or current regulation
3	VOUT_ADC_EN	1	R/W	0: disable VOUT sampling 1: enable VOUT sampling Selection does not influence VOUT protections and/or regulation
2	IOUT_ADC_EN	1	R/W	0: disable IOUT sampling 1: enable IOUT sampling Selection does not influence IOUT protections and/or regulation current regulation
1	OTP_ADC_EN	1	R/W	0: disable OTP sampling 1: enable OTP sampling Selection does not influence temperature protections
0	NTC_ADC_EN	1	R/W	0: disable NTC sampling 1: enable NTC sampling Selection does not influence temperature protections

8.13.3.15 VPWR_CONFIG Register (Address 0Fh)

The VPWR configuration register.

This is a READ/WRITE register.

Table 21. VPWR_CONFIG Register

Bit	Symbol	Default Value	Type	Function
7	VPWR_EN	1	R/W	VPWR enable bit: 0: disable VPWR LDO output 1: Enable VPWR LDO output
6	RSVD	0	R/W	Reserved
5	RSVD	0	R/W	Reserved

Table 21. VPWR_CONFIG Register...continued

Bit	Symbol	Default Value	Type	Function
4	RSVD	0	R/W	Reserved
3	RSVD	0	R/W	Reserved
2	RSVD	0	R/W	Reserved
1	VPWR_VL[1]	11	R/W	VPWR output voltage setting bits: 00: VPWR=1.8 V 01: VPWR=2.5 V 10: VPWR=2.8 V 11: VPWR=3.3 V (Default)
0	VPWR_VL[0]			

8.13.3.16 RXIR_CONFIG Register (Address 10h)

The RXIR function configuration register.

This is a READ/WRITE register.

Table 22. RXIR_CONFIG Register

Bit	Symbol	Default Value	Type	Function
7	RXIR_EN	0	R/W	RXIR enable bit: 0: RXIR function is disabled 1: RXIR function is enabled The bit is cleared by PCA9430 when RXIR is finished
6	RSVD	0	R/W	Reserved
5	RXIR_DUR[2:0]	100	R/W	RXIR pull down duration selection bits: 000: 0.2 ms 001: 0.4 ms 010: 0.6 ms 011: 0.8 ms 100: 1.0 ms (default) 101: 1.2 ms 110: 1.4 ms 111: 1.6 ms
4				
3				
2	RXIR_STR[2:0]	000	R/W	RXIR pull down resistor on AC1 and AC2 selection bits: 000: 20 Ω (default) 001: 80 Ω 010: 140 Ω 011: 200 Ω 100: 260 Ω 101: 320 Ω 110: 380 Ω 111: 440 Ω
1				
0				

8.13.3.17 BQ_CONTROL Register (Address 20h)

This register stores the linear battery charge related control registers.

This is a READ AND WRITE register.

Table 23. BQ_CONTROL Register

Bit	Symbol	Default Value	Type	Function
7	CHG_LOCK[4:0]	10101	R/W	Critical charger related setting lock. CHG_LOCK[4:0] = 10101, these registers which are labeled as "locked by CHG_LOCK" can be accessed to perform I ² C "write" command. CHGN_LOCK[4:0] ≠ 10101, these registers which are labeled as "locked by CHG_LOCK" can NOT be accessed to perform I ² C "write" command. In such case when "write" command is performed on these locked registers, it returns the present register value.
6				
5				
4				
3				
2	NTC_EN	0	R/W	Enable TS pin external thermistor (NTC) control in charger: 0: Thermistor (NTC) control disabled in charger 1: Thermistor (NTC) control active in charger
1	CHG_TIMER_EN	1	R/W	Enable the fast charge timer and pre-qual timer: 0: fast charge timer and pre-qual timer are both disabled 1: fast charge timer and pre-qual timer are both enabled
0	RSVD	0	R/W	Reserved

8.13.3.18 CC_CURRENT Register (Address 21h)

The CC_CURRENT register is CC (constant current) state charging current setting for host.

This is a READ AND WRITE register. This register is locked by CHG_LOCK.

Table 24. CC_CURRENT Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	RSVD	0	R/W	Reserved
5	ICHG_CC[5:0]	001000	R/W	Linear battery charger constant current (CC) setting: 00h: 0 mA 10h: 80 mA 20h: 160 mA 30h: 240 mA 01h: 5 mA 11h: 85 mA 21h: 165 mA 31h: 245 mA 02h: 10 mA 12h: 90 mA 22h: 170 mA 32h: 250 mA 03h: 15 mA 13h: 95 mA 23h: 175 mA 33h: 255 mA 04h: 20 mA 14h: 100 mA 24h: 180 mA 34h: 260 mA 05h: 25 mA 15h: 105 mA 25h: 185 mA 35h: 265 mA 06h: 30 mA 16h: 110 mA 26h: 190 mA 36h: 270 mA 07h: 35 mA 17h: 115 mA 27h: 195 mA 37h: 275 mA 08h: 40 mA (default) 18h: 120 mA 28h: 200 mA 38h: 280 mA 09h: 45 mA 19h: 125 mA 29h: 205 mA 39h: 285 mA 0Ah: 50 mA 1Ah: 130 mA 2Ah: 210 mA 3Ah: 290 mA 0Bh: 55 mA 1Bh: 135 mA 2Bh: 215 mA 3Bh: 295 mA 0Ch: 60 mA 1Ch: 140 mA 2Ch: 220 mA 3Ch: 300 mA 0Dh: 65 mA 1Dh: 145 mA 2Dh: 225 mA 3Dh: 305 mA 0Eh: 70 mA 1Eh: 150 mA 2Eh: 230 mA 3Eh: 310 mA 0Fh: 75 mA 1Fh: 155 mA 2Fh: 235 mA 3Fh: 315 mA
4				
3				
2				
1				
0				

8.13.3.19 TP_CURRENT Register (Address 22h)

The TP_CURRENT register is Top-off state charging current setting for host.

This is a READ AND WRITE register, and this register is locked by CHG_LOCK.

Table 25. TP_CURRENT Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	RSVD	0	R/W	Reserved
5	ICHG_TOPOFF [5:0]	000100b	R/W	Linear battery charger top-off charge current setting: 00h: 0 mA 10h: 16 mA 20h: 32 mA 30h: 48 mA 01h: 1 mA 11h: 17 mA 21h: 33 mA 31h: 49 mA 02h: 2 mA 12h: 18 mA 22h: 34 mA 32h: 50 mA 03h: 3 mA 13h: 19 mA 23h: 35 mA 33h: 51 mA 04h: 4 mA (default) 14h: 20 mA 24h: 36 mA 34h: 52 mA 05h: 5 mA 15h: 21 mA 25h: 37 mA 35h: 53 mA 06h: 6 mA 16h: 22 mA 26h: 38 mA 36h: 54 mA 07h: 7 mA 17h: 23 mA 27h: 39 mA 37h: 55 mA 08h: 8 mA 18h: 24 mA 28h: 40 mA 38h: 56 mA 09h: 9 mA 19h: 25 mA 29h: 41 mA 39h: 57 mA 0Ah: 10 mA 1Ah: 26 mA 2Ah: 42 mA 3Ah: 58 mA 0Bh: 11 mA 1Bh: 27 mA 2Bh: 43 mA 3Bh: 59 mA 0Ch: 12 mA 1Ch: 28 mA 2Ch: 44 mA 3Ch: 60 mA 0Dh: 13 mA 1Dh: 29 mA 2Dh: 45 mA 3Dh: 61 mA 0Eh: 14 mA 1Eh: 30 mA 2Eh: 46 mA 3Eh: 62 mA 0Fh: 15 mA 1Fh: 31 mA 2Fh: 47 mA 3Fh: 63 mA
4				
3				
2				
1				
0				

8.13.3.20 LB_CURRENT Register (Address 23h)

The LB_CURRENT register is low battery state charging current setting for host.

This is a READ AND WRITE register, and this register is locked by CHG_LOCK.

Table 26. LB_CURRENT Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	RSVD	0	R/W	Reserved
5	ICHG_LOW [5:0]	001000b	R/W	Low battery charge current setting: 00h: 0 mA 10h: 16 mA 20h: 32 mA 30h: 48 mA 01h: 1 mA 11h: 17 mA 21h: 33 mA 31h: 49 mA 02h: 2 mA 12h: 18 mA 22h: 34 mA 32h: 50 mA 03h: 3 mA 13h: 19 mA 23h: 35 mA 33h: 51 mA 04h: 4 mA 14h: 20 mA 24h: 36 mA 34h: 52 mA 05h: 5 mA 15h: 21 mA 25h: 37 mA 35h: 53 mA 06h: 6 mA 16h: 22 mA 26h: 38 mA 36h: 54 mA 07h: 7 mA 17h: 23 mA 27h: 39 mA 37h: 55 mA 08h: 8 mA (default) 18h: 24 mA 28h: 40 mA 38h: 56 mA 09h: 9 mA 19h: 25 mA 29h: 41 mA 39h: 57 mA 0Ah: 10 mA 1Ah: 26 mA 2Ah: 42 mA 3Ah: 58 mA 0Bh: 11 mA 1Bh: 27 mA 2Bh: 43 mA 3Bh: 59 mA 0Ch: 12 mA 1Ch: 28 mA 2Ch: 44 mA 3Ch: 60 mA 0Dh: 13 mA 1Dh: 29 mA 2Dh: 45 mA 3Dh: 61 mA 0Eh: 14 mA 1Eh: 30 mA 2Eh: 46 mA 3Eh: 62 mA 0Fh: 15 mA 1Fh: 31 mA 2Fh: 47 mA 3Fh: 63 mA
4				
3				
2				
1				
0				

8.13.3.21 DB_CURRENT Register (Address 24h)

The DB_CURRENT register is dead battery state charging current setting for host.

This is a READ AND WRITE register, and this register is locked by CHG_LOCK.

Table 27. DB_CURRENT Register

Bit	Symbol	Default Value	Type	Function
7	ICHG_DEAD_TIMER	01	R/W	Linear battery charger dead charge timer setting for battery short detection, 00: 5s; 01: 10s; 10: 20s; 11: 40s
6				
5	ICHG_DEAD [5:0]	000100b	R/W	Dead battery charge current setting: 00h: 0 mA 10h: 16 mA 20h: 32 mA 30h: 48 mA 01h: 1 mA 11h: 17 mA 21h: 33 mA 31h: 49 mA 02h: 2 mA 12h: 18 mA 22h: 34 mA 32h: 50 mA 03h: 3 mA 13h: 19 mA 23h: 35 mA 33h: 51 mA 04h: 4 mA (default) 14h: 20 mA 24h: 36 mA 34h: 52 mA 05h: 5 mA 15h: 21 mA 25h: 37 mA 35h: 53 mA 06h: 6 mA 16h: 22 mA 26h: 38 mA 36h: 54 mA 07h: 7 mA 17h: 23 mA 27h: 39 mA 37h: 55 mA 08h: 8 mA 18h: 24 mA 28h: 40 mA 38h: 56 mA 09h: 9 mA 19h: 25 mA 29h: 41 mA 39h: 57 mA 0Ah: 10 mA 1Ah: 26 mA 2Ah: 42 mA 3Ah: 58 mA 0Bh: 11 mA 1Bh: 27 mA 2Bh: 43 mA 3Bh: 59 mA 0Ch: 12 mA 1Ch: 28 mA 2Ch: 44 mA 3Ch: 60 mA 0Dh: 13 mA 1Dh: 29 mA 2Dh: 45 mA 3Dh: 61 mA 0Eh: 14 mA 1Eh: 30 mA 2Eh: 46 mA 3Eh: 62 mA 0Fh: 15 mA 1Fh: 31 mA 2Fh: 47 mA 3Fh: 63 mA Note: Current value set in ICHG_DEAD[5:0] should NOT be greater than the value set in ICHG_LOW[5:0]. Note: The current default value for ICHG_DEAD[5:0] is set at 4 mA, ICHG_DEAD[2] and ICHG_DEAD[4] should be MTP programmable.
4				
3				
2				
1				
0				

8.13.3.22 VBAT_REG Register (Address 25h)

This register set the regulated battery voltage and restart battery voltage threshold.

This is a READ AND WRITE register, and this register is locked by CHG_LOCK.

Table 28. VBAT_REG Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R/W	Reserved
6	VBAT_RESTART	0	R/W	VBAT Restart charge threshold: 0: 140 mV; 1: 240 mV

Table 28. VBAT_REG Register...continued

Bit	Symbol	Default Value	Type	Function
5	VBAT_REG[5:0]	011110b	R/W	VBATREG, linear battery charger regulated battery voltage setting:
4				00h: 3.60 V 10h: 3.92 V 20h: 4.24 V 30h: 4.56 V
3				01h: 3.62 V 11h: 3.94 V 21h: 4.26 V 31h: 4.58 V
2				02h: 3.64 V 12h: 3.96 V 22h: 4.28 V 32h: 4.60 V
1				03h: 3.66 V 13h: 3.98 V 23h: 4.30 V 33~3Fh: 4.60 V
0				04h: 3.68 V 14h: 4.00 V 24h: 4.32 V
				05h: 3.70 V 15h: 4.02 V 25h: 4.34 V
				06h: 3.72 V 16h: 4.04 V 26h: 4.36 V
				07h: 3.74 V 17h: 4.06 V 27h: 4.38 V
				08h: 3.76 V 18h: 4.08 V 28h: 4.40 V
				09h: 3.78 V 19h: 4.10 V 29h: 4.42 V
				0Ah: 3.80 V 1Ah: 4.12 V 2Ah: 4.44 V
				0Bh: 3.82 V 1Bh: 4.14 V 2Bh: 4.46 V
				0Ch: 3.84 V 1Ch: 4.16 V 2Ch: 4.48 V
				0Dh: 3.86 V 1Dh: 4.18 V 2Dh: 4.50 V
				0Eh: 3.88 V 1Eh: 4.20 V (Default) 2Eh: 4.52 V
				0Fh: 3.90 V 1Fh: 4.22 V 2Fh: 4.54 V

8.13.3.23 BQTIMER_AND_NTC Register (Address 26h)

This register set the linear battery charger timer and external NTC thermistor resistance.

This is a READ AND WRITE register, and this register is locked by CHG_LOCK.

Table 29. BQTIMER_AND_NTC Register

Bit	Symbol	Default Value	Type	Function
7	NTC_RES_SEL	1	R/W	External thermistor typical resistance selection: 0: 100 kΩ; 1: 10 kΩ
6	TIMER_2X	0	R/W	Charging Safety Timer Extension: 0: Both pre-qual and fast charge timer duration keeps as the values set in ICHG_PREQ_TIMER[1:0] and ICHG_FAST_TIMER[1:0] 1: Both pre-qual and fast charge timer duration is extended to 2x of the values set in ICHG_PREQ_TIMER[1:0] and ICHG_FAST_TIMER[1:0]
5	ICHG_FAST_TIMER[1:0]	01	R/W	Linear battery charger fast charge timer setting: 00: 3 hr; 01: 5 hr; 10: 7 hr; 11: 9 hr
4				
3	ICHG_PREQ_TIMER[1:0]	01	R/W	Linear battery charger pre-qualification charge timer setting: 00: 15 min; 01:30 min; 10: 45 min; 11: 60 min
2				
1	T_TOPOFF[1:0]	01	R/W	TOPOFF Timer setting: 00: 0 min; 01:6.4 min; 10: 12.8 min; 11: 19.2 min
0				

8.13.3.24 BQTHM_AND_SHRT Register (Address 27h)

This register stores the NTC Beta value selection, VBAT short circuit detection and thermal regulation setting.

This is a READ AND WRITE register.

Table 30. BQTHM_AND_SHRT Register

Bit	Symbol	Default Value	Type	Function
7	NTC_BETA_SEL[2:0]	001	R/W	Set the thermistor beta value selection: 000: 3435k 010: 3934k 100: 4100k 110: 4543k 001: 3610k 011: 3950k 101: 4311k 111: 4750k
6				
5				
4	RSVD	0	R/W	Reserved
3	RSVD	0	R/W	Reserved
2	THM_REG[2:0]	100	R/W	Thermal regulation threshold setting: 000: 90 °C 010: 100 °C 100: 110 °C 110: 120 °C 001: 95 °C 011: 105 °C 101: 115 °C 111: 125 °C
1				
0				

8.13.3.25 Battery Charger Status_1 Register (Address 28h)

This register stores the present status of the linear battery charger, part 1.

This is a READ ONLY register.

Table 31. Battery Charger Status_1 Register

Bit	Symbol	Default Value	Type	Function
7	VBAT_DET_OK	0	R	VBAT Detection Status, the bit is reset when CHG_EN=1'b0: 0: No valid battery attachment detected. 1: Battery attachment detected.
6	VBAT_OK	0	R	VBAT status, refer to BAT_DETAIL_STATUS [2:0] for more details. The bit is reset when CHG_EN=1'b0: 0: the battery is invalid/missing, or charger reset is active, i.e. BAT_DETAIL_STATUS[2:0] = 0b000, 0b111 1: the battery is OK. i.e. BAT_DETAIL_STATUS[2:0] = 0b010, 0b011, 0b100, 0b101
5	VRECT_OK	0	R	VRECT status, refer to VRECT_STATUS [1:0] for more details 0: VRECT voltage is invalid. i.e. VRECT_STATUS[1:0] ≠ 0b11 1: VRECT voltage is valid. i.e. VRECT_STATUS[1:0] = 0b11
4	CHG_OK	0	R	Charger status, the bit is reset when CHG_EN=1'b0: 0: The charger has suspended due to the following conditions: TS_DETAIL_STATUS[2:0] = 001'b or 100'b; or SFTY_TIMER_STATUS[1:0] ≠ 00'b 1: The charger is OK
3	RSVD	0	R	Reserved

Table 31. Battery Charger Status_1 Register...continued

Bit	Symbol	Default Value	Type	Function
2	RSVD	0	R	Reserved
1	RSVD	0	R	Reserved
0	RSVD	0	R	Reserved

8.13.3.26 Battery Charger Status_2 Register (Address 29h)

This register stores the present status of the linear battery charger, part 2.

This is a READ ONLY register.

Table 32. Battery Charger Status_2 Register

Bit	Symbol	Default Value	Type	Function
7	VRECT_STATUS	00	R	VRECT Status: 00: VRECT is invalid. $V_{RECT} < V_{RECT_UVLO}$ 01: VRECT is invalid. $V_{RECT} < V_{BAT} + V_{RECT2BAT_HEADROOM}$ and $V_{RECT} > V_{RECT_UVLO}$ 10: VRECT is invalid. $V_{RECT} > V_{RECT_OVP}$ 11: VRECT is valid. $V_{RECT} > V_{RECT_UVLO}$, $V_{RECT} > V_{BAT} + V_{RECT2BAT_HEADROOM}$, $V_{RECT} < V_{RECT_OVP}$
6				
5	RSVD	0	R	Reserved
4	TREG_STATUS	0	R	Temperature Regulation Loop Status, The bit is reset when $CHG_EN = 1'b0$: 0: Die junction temperature is less than the threshold set by THM_REG and the full charge current limit is available. 1: Die junction temperature is greater than the threshold set by THM_REG, and the charge current limit may be folding back to reduce power dissipation.
3	RSVD	0	R	Reserved
2	RSVD	0	R	Reserved
1	RSVD	0	R	Reserved
0	RSVD	0	R	Reserved

8.13.3.27 Battery Charger Status_3 Register (Address 2Ah)

This register stores the present status of the linear battery charger, part 3.

This is a READ ONLY register.

Table 33. Battery Charger Status_3 Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R	Reserved

Table 33. Battery Charger Status_3 Register...continued

Bit	Symbol	Default Value	Type	Function
6	BAT_DETAIL_STATUS[2:0]	111	R	Battery Details, the bits are reset when CHG_EN=1'b0: 000: Battery missing, not attached. 001: Battery detection in-progress 010: Battery is valid, VBAT < Vbat_dead level 011: Battery is valid, Vbat_dead level < VBAT < V _{BAT_LOW} level 100: Battery is valid, V _{BAT_LOW} level < VBAT < Vbat_reg – Vbat_restart 101: Battery is valid, VBAT > Vbat_reg – Vbat_restart 110: reserved 111: battery charger is in reset
5				
4				
3	RSVD	0	R	Reserved
2	BAT_CHG_STATUS[2:0]	000	R	The bits are reset when CHG_EN=1'b0, 000: Charger in Idle State 001: Charger in Dead-Battery State 010: Charger in Low-Battery State 011: Charger in Fast Charging State (CC) 100: Charger in Fast Charging State (CV) 101: Charger in Top-off State 110: Charger in Done State 111: Reserved
1				
0				

8.13.3.28 Battery Charger Status_4 Register (Address 2Bh)

This register stores the present status of the linear battery charger, part 4.

This is a READ ONLY register.

Table 34. Battery Charger Status_4 Register

Bit	Symbol	Default Value	Type	Function
7	RSVD	0	R	Reserved
6	TS_DETAIL_STATUS[2:0]	000	R	The bits are reset when CHG_EN=1'b0, 000: Battery Temperature Nominal, $T_2 \leq T \leq T_3$ 001: Battery Temperature is Cold, $T < T_1$ 010: Battery Temperature is Cool, $T_1 \leq T < T_2$ 011: Battery Temperature is Warm, $T_3 < T \leq T_4$ 100: Battery Temperature is Hot, $T > T_4$
5				
4				
3	CHIP_TEMP_STATUS[1:0]	00	R	The bits are reset when CHG_EN=1'b0, 00: Chip Temperature having No effect on Battery Charging 01: Chip Temperature above Regulation Threshold, below Shutdown Threshold 10: Chip Temperature above Shutdown Threshold 11: Reserved
2				
1	SFTY_TIMER_STATUS[1:0]	00	R	00: Safety Timers having No Effect on Battery Charging 01: Pre-qual Timer expires, battery charging suspended 10: Fast Timer expires, battery Charging suspended 11: Battery short test fails, battery charging suspended
0				

8.13.3.29 VRECT_ ADC Register (Address 30h, 31h)

The register contains VRECT voltage ADC measurement results.

This is a READ ONLY register.

Table 35. VRECT_ ADC Register

Address	Bit	Symbol	Default Value	Type	Function
30h	15	VRECT[9:0]	00h	R	VRECT voltage resolution is LSB:15.82 mV
	14				
	13				
	12				
	11				
	10				
	9				
	8				
31h	7	RSVD	0	R	Reserved
	6				
	5				
	4				
	3				
	2				
	1				
	0				

8.13.3.30 VTUNE_ ADC Register (Address 32h, 33h)

The register contains VTUNE voltage ADC measurement results.

This is a READ ONLY register.

Table 36. VTUNE_ ADC Register

Address	Bit	Symbol	Default Value	Type	Function
32h	15	VTUNE[9:0]	00h	R	VTUNE voltage resolution is LSB:3.32 mV
	14				
	13				
	12				
	11				
	10				
	9				
	8				
33h	7	RSVD	0	R	Reserved
	6				
	5				

Table 36. VTUNE_ ADC Register...continued

Address	Bit	Symbol	Default Value	Type	Function
	4	RSVD	0	R	Reserved
	3	RSVD	0	R	Reserved
	2	RSVD	0	R	Reserved
	1	RSVD	0	R	Reserved
	0	RSVD	0	R	Reserved

8.13.3.31 VOUT_ ADC Register (Address 34h, 35h)

The register contains VOUT voltage ADC measurement results.

This is a READ ONLY register.

Table 37. VOUT_ ADC Register

Address	Bit	Symbol	Default Value	Type	Function
34h	15	VOUT[9:0]	00h	R	VOUT voltage resolution is LSB: 5.27 mV
	14				
	13				
	12				
	11				
	10				
	9				
	8				
35h	7				
	6				
	5	RSVD	0	R	Reserved
	4	RSVD	0	R	Reserved
	3	RSVD	0	R	Reserved
	2	RSVD	0	R	Reserved
	1	RSVD	0	R	Reserved
	0	RSVD	0	R	Reserved

8.13.3.32 IOUT_ ADC Register (Address 32h, 33h)

The register contains IOUT voltage ADC measurement results.

This is a READ ONLY register.

Table 38. IOUT_ ADC Register

Address	Bit	Symbol	Default Value	Type	Function
36h	15	IOUT[9:0]	00h	R	IOUT voltage resolution is LSB: 421.9 μ A
	14				
	13				

Table 38. IOUT_ADC Register...continued

Address	Bit	Symbol	Default Value	Type	Function
37h	12				
	11				
	10				
	9				
	8				
	7				
	6				
	5	RSVD	0	R	Reserved
	4	RSVD	0	R	Reserved
	3	RSVD	0	R	Reserved
	2	RSVD	0	R	Reserved
	1	RSVD	0	R	Reserved
	0	RSVD	0	R	Reserved

8.13.3.33 IRECT_ADC Register (Address 38h, 39h)

The register contains IOUT voltage ADC measurement results.

This is a READ ONLY register.

Table 39. IRECT_ADC Register

Address	Bit	Symbol	Default Value	Type	Function
38h	15	IRECT[9:0]	00h	R	IRECT current resolution is LSB: 421.9 μ A. IRECT current is the sum of VOUT and VPWR load current. In application, the operation current can be added to calculate the received power.
	14				
	13				
	12				
	11				
	10				
	9				
	8				
39h	7				
	6				
	5	RSVD	0	R	Reserved
	4	RSVD	0	R	Reserved
	3	RSVD	0	R	Reserved
	2	RSVD	0	R	Reserved
	1	RSVD	0	R	Reserved
	0	RSVD	0	R	Reserved

8.13.3.34 TDIE_ ADC Register (Address 3Ah, 3Bh)

The register contains TDIE ADC measurement results.

This is a READ ONLY register.

Table 40. TDIE_ ADC Register

Address	Bit	Symbol	Default Value	Type	Function
3Ah	7	TDIE[7:0]	00h	R	On chip temperature ADC measurement results: 00h: -43 °C 10h: 27 °C 20h: 96 °C 30h: 159 °C 01h: -39 °C 11h: 31 °C 21h: 100.5 °C 31h: 163 °C 02h: -35 °C 12h: 35 °C 22h: 105 °C 32h - FFh: Reserved 03h: -30 °C 13h: 40 °C 23h: 109 °C 04h: -26 °C 14h: 44 °C 24h: 114 °C 05h: -21 °C 15h: 48 °C 25h: 118 °C 06h: -17 °C 16h: 53 °C 26h: 122 °C 07h: -12.5 °C 17h: 57 °C 27h: 126 °C 08h: -8 °C 18h: 62 °C 28h: 130 °C 09h: -4 °C 19h: 65.5 °C 29h: 134.5 °C 0Ah: 0 °C 1Ah: 71 °C 2Ah: 138 °C 0Bh: 5 °C 1Bh: 75 °C 2Bh: 141 °C 0Ch: 9 °C 1Ch: 79.5 °C 2Ch: 145 °C 0Dh: 13.5 °C 1Dh: 84 °C 2Dh: 148 °C 0Eh: 18 °C 1Eh: 88 °C 2Eh: 152 °C 0Fh: 22.5 °C 1Fh: 92 °C 2Fh: 156 °C
	6				
	5				
	4				
	3				
	2				
	1				
	0				

8.13.3.35 NTC_ ADC Register (Address 3Ch, 3Dh)

The register contains NTC voltage ADC measurement results.

This is a READ ONLY register.

Table 41. NTC_ ADC Register

Address	Bit	Symbol	Default value	Type	Function
3Bh	15	NTC[9:0]	00h	R	NTC voltage resolution is LSB: 2.64 mV
	14				
	13				
	12				
	11				
	10				
	9				
	8				
3Ch	7	RSVD	0	R	Reserved
	6				
	5				
	4				

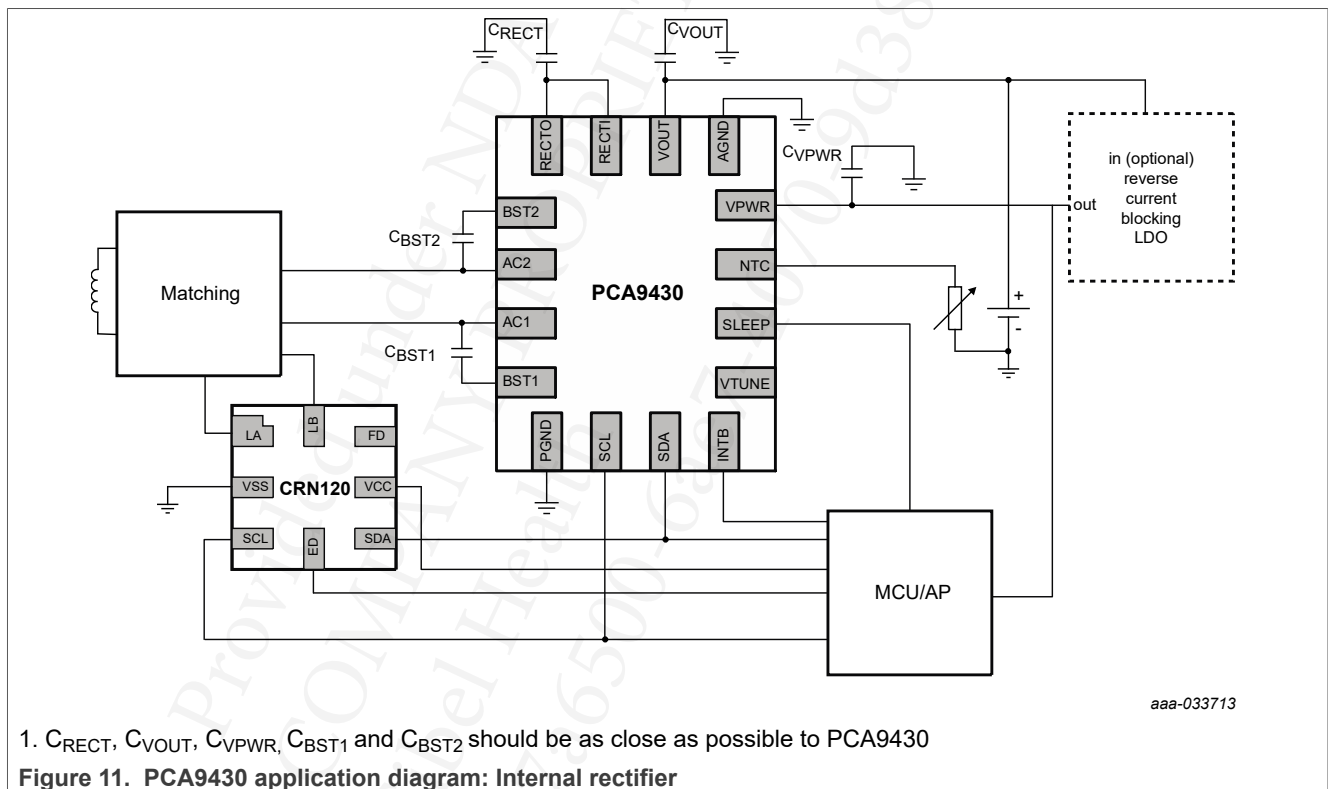
Table 41. NTC_ADC Register...continued

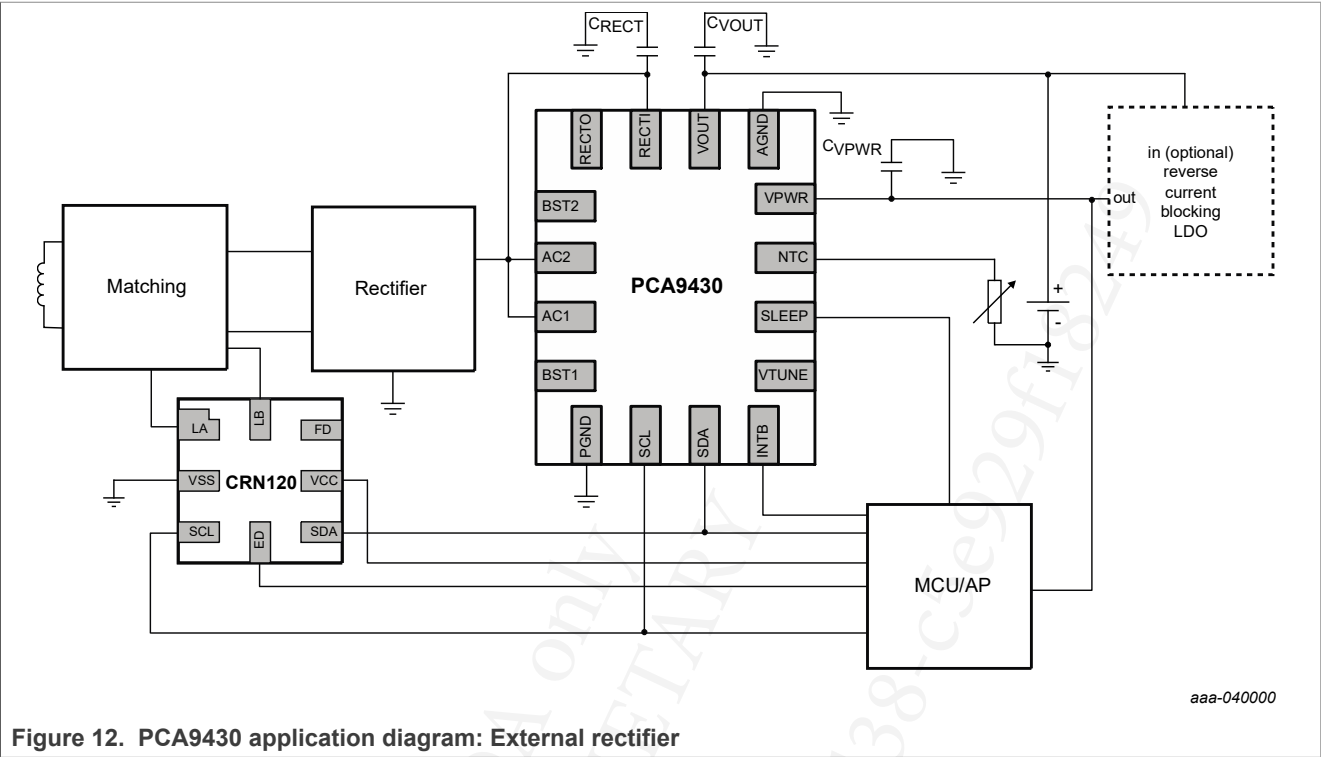
Address	Bit	Symbol	Default value	Type	Function
	3	RSVD	0	R	Reserved
	2	RSVD	0	R	Reserved
	1	RSVD	0	R	Reserved
	0	RSVD	0	R	Reserved

9 Application diagram

The PCA9430 device is typically used in wearable and IOT application as NFC wireless charger. It shares the same NFC coil so there is no additional cost. It is recommended to add a varactor, which can be tuned to have best resonance to take power.

The device output is a linear battery charger, which is connected to 1-cell lion battery directly as shown in Figure 11. In addition, VPWR LDO provides power to system MCU or AP at dead battery condition; that enables system MCU or AP wake up and communication with NFC transmitter to negotiate for high power. The LDO has reverse current protection (RCP) feature and can be connected directly to output of system LDO with reverse current blocking feature.





10 Limiting values

Table 42. Limiting values

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	Min	Max	Unit
V _I	input voltage	AC1, AC2	-0.5	+18	V
		NTC	-0.5	+7.0	V
		SCL, SDA, SLEEP	-0.5	+7.0	V
		RECTI	-0.5	+18	V
V _O	output voltage	BST1; reference to AC1	AC1	AC1+5	V
		BST2; reference to AC2	AC2	AC2+5	V
		BST1; reference to Ground	-0.5	+7.0	V
		BST2; reference to Ground	-0.5	+7.0	V
V _O	output voltage	RECTO	-0.5	+18	V
		VOUT; VPWR	-0.5	+7.0	V
		INTB	-0.5	+7.0	V
		AVDD; VTUNE	-0.5	+7.0	V
I _I	Input current	AC1, AC2, RECTI		300	mA
I _O	Output current	VOUT		300	mA
		VPWR, RECTO		50	mA
I _{IK}	input clamping current	V _I < -0.5 V	-50	-	mA
I _{SK}	switch clamping current	AVDD, VOUT, VTUNE, VPWR; V _I < -0.5 V	-50	-	mA
T _{stg}	storage temperature		-65	+150	°C
P _{tot} ^[1]	total power dissipation		-	1.4	W

[1] The (absolute) maximum power dissipation depends on the junction temperature T_j. Higher power dissipation is allowed in conjunction with lower ambient temperatures. The conditions to determine the specified values are T_{amb} = 25 °C and the use of a 4-layer PCB.

11 Recommended operating conditions

Table 43. Recommended operating conditions

Symbol	Parameter	Conditions	Min	Max	Unit
V _I	input voltage	AC1, AC2	3	13	V
		RECTI	3	12	V
		SCL, SDA, NTC	0	3.3	V
		SLEEP	0	5	V
V _O	output voltage	VPWR	0	3.3	V
		VOUT	0	5	V
		VTUNE	0	3.3	V
		RECTO	3	12	V
I _I	Input current	RECTI		200	mA
		AC1, AC2, (using external rectifier)		200	mA
		AC1, AC2, (using internal rectifier)		40	mA
I _O	Output current	VOUT		200	mA
		RECTO		40	mA
C	Capacitance	C _{BST1} , C _{BST2}	22	50	nF
		C _{RECTO}	10		μF
		C _{PWR}	1	4.7	μF
		C _{OUT}	4.7		μF
T _{j(max)}	maximum junction temperature		-40	+125	°C
T _{amb}	ambient temperature		-40	+85	°C

12 Thermal characteristics

Table 44. Thermal characteristics

Symbol	Parameter	Conditions	Typ	Unit
R _{th(j-a)}	thermal resistance from junction to ambient		[1][2] 67.4	K/W

[1] The overall R_{th(j-a)} can vary depending on the board layout. To minimize the effective R_{th(j-a)}, all pins must have a solid connection to larger Cu layer areas e.g. to the power and ground layer. In multi-layer PCB applications, the second layer should be used to create a large heat spreader area right below the device. If this layer is either ground or power, it should be connected with several vias to the top layer connecting to the device ground or supply. Try not to use any solder-stop varnish under the chip.

[2] This R_{th(j-a)} is calculated based on JEDEX2S2P board. The actual R_{th(j-a)} value may vary in applications using different layer stacks and layouts.

13 Static characteristics

Table 45. General characteristics

At recommended operating conditions; voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	T _{amb} = 25 °C			T _{amb} = -40 °C to +85 °C		Unit
			Min	Typ	Max	Min	Max	
VRECT Specifications								
I _{SLEEP}	VRECT sleep mode current	VRECT = 5 V, the device is in sleep mode	-	31	-	-	63	μA
I _{VRECT_STATIC}	VRECT static current	VRECT = 5 V, both battery charger and VPWR are off	-	317	-	-	600	μA
		VRECT = 5 V, battery charger is off and VPWR is on, I _{VPWR} = 0	-	333	-	-	600	μA
		VRECT = 5 V, battery charger is on and VPWR is off. VOUT = VBAT_REG + 200 mV	-	1.04	-	-	2	mA
		VRECT = 5V, both battery charger and VPWR are on. VOUT = VBAT_REG + 200 mV and I _{VPWR} = 0	-	1.06	-	-	2	mA
V _{RECT_UVLO}	VRECT undervoltage lockout release voltage	VRECT Rising	-	3.0	-	2.85	3.15	V
V _{RECT_UVLO_AC}	UVLO trip level accuracy	When the UVLO threshold is set to the other values by I ² C register	-	-	-	-3.5	3.5	%
V _{hys(UVLO)}	VRECT undervoltage lockout hysteresis voltage	VRECT Falling	-	100	-			mV
V _{RECT_OVP}	VRECT overvoltage protection voltage	VRECT Rising	-	14	-	13.5	14.5	V
		VRECT Rising; When device is in SLEEP mode	-	15.4	-	14.5	16.3	V
V _{RECT_OVP_AC}	VRECT OVP trip level accuracy	When the OVP threshold is set to the other values by I ² C register	-	-	-	-3.5	3.5	%
V _{hys(OVP)}	VRECT Overvoltage lockout hysteresis voltage	VRECT Falling	-	1	-			V
t _{OVP}	VRECT Overvoltage protection response time	VRECT rising and ramp up slew rate is 100 mV/μs	-	2	-		3	μs
V _{RECT_REGLOW_DELTA}	The delta between VRECT low regulation voltage and VBAT	When I _{CHG} = I _{CHG_CC} × 5 %	-	220	-	180	260	mV
V _{RECT_REGLOW_MIN}	VRECT minimum low regulation voltage		-	3.6	-	3.45	3.75	V
V _{RECT_REGHIGH}	VRECT HIGH regulation voltage		-	12	-	11.58	12.42	V

Table 45. General characteristics...continued

At recommended operating conditions; voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	T _{amb} = 25 °C			T _{amb} = -40 °C to +85 °C		Unit
			Min	Typ	Max	Min	Max	
V _{RECT_REGHIGH_AC}	VRECT HIGH regulation voltage accuracy	When the REGHIGH threshold is set to the other values by I ² C register	-	-	-	-3.5	3.5	%
V _{RECT_OVWP}	VRECT pre-charging overvoltage warning voltage		-	7	-	6.7	7.3	V
V _{RECT_OVWP_AC}	VRECT overvoltage warning voltage accuracy	When the OVWP threshold is set to the other values by I ² C register	-	-	-	-3.5	3.5	%
V _{RECT_OVFAST_MIN}	VRECT fast charging overvoltage warning min voltage	VRECT Rising	-	4	-	3.8	4.2	V
V _{RECT_OVFAST_MIN_AC}	VRECT fast charging overvoltage warning min voltage accuracy		-	-	-	-3.5	3.5	%
V _{RECT_OVFAST_DELTA}	VRECT fast charging overvoltage warning delta voltage	VRECT Rising	-	0.75	-	0.7	0.82	V
V _{RECT_OVFAST_DELTA_AC}	VRECT fast charging overvoltage warning delta voltage accuracy	When the OVFAST threshold is set to the other values by I ² C register	-	-	-	-10	10	%
V _{RECT2BAT_HEADROOM}	VRECT to VBAT headroom		-	100	-	75	125	mV
Timer Specifications								
t _{WD_TIMER_RANGE}	Watchdog timer	Watchdog timer range	-	-	-	16	64	s
t _{WD_TIMER_AC}	Watchdog timer	Watchdog timer accuracy	-	-	-	-15	15	%
t _{CHG_PREQ_RANGE}	Pre-charging safety timer	Pre-charging safety timer range	-	-	-	15	60	min
t _{CHG_PREQ_AC}	Pre-charging safety timer	Pre-charging safety timer accuracy	-	-	-	-15	15	%
t _{CHG_FAST_RANGE}	Fast charging safety timer	Fast charging safety timer range	-	-	-	3	9	hr
t _{CHG_FAST_AC}	Fast charging safety timer	Fast charging safety timer accuracy	-	-	-	-15	15	%
SLEEP pin Specifications								
V _{IH}	HIGH-level input voltage	SLEEP	1.2	-	-	1.2	-	V
V _{IL}	LOW-level input voltage	SLEEP	-	-	0.4	-	0.4	V
t _{SLEEP_MIN_WIDTH}	Minimum pulse width	SLEEP; For both logic HIGH and logic LOW	50	-	-	50	-	μs
I²C-bus interface Specifications								
V _{IH}	HIGH-level input voltage	SCL, SDA	1.2	-	-	1.2	-	V
V _{IL}	LOW-level input voltage	SCL, SDA	-	-	0.4	-	0.4	V

Table 45. General characteristics...continued

At recommended operating conditions; voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	T _{amb} = 25 °C			T _{amb} = -40 °C to +85 °C		Unit
			Min	Typ	Max	Min	Max	
V _{OL}	LOW-level output voltage	INTB pin; I _{load} = 1 mA	-	-	0.4	-	0.4	V
f _{CLK_I2C}	I ² C bus clock frequency		0	-	1	0	1	MHz
VTUNE Specifications								
VTUNE	5-bit DAC output voltage range		-	-	-	0	3.3	V
V _{LSB}	1 LSB voltage		-	103	-	99	107	mV
INL	Integral Nonlinearity		-1	-	1	-1	1	LSB
DNL	Differential Nonlinearity		-0.5	-	0.5	-0.5	0.5	LSB
ADC Specifications								
Nres(ADC)	ADC Resolution		-	10	-	-	-	bit
f _s	Sampling frequency		-	91	-	-	-	ksps
VRECT_ADC	VRECT ADC monitor range		0	-	16.2	0	16.2	V
	VRECT ADC monitor effective resolution		-	15.82	-	-	-	mV
	VRECT ADC monitor tolerance	VRECT = 5.0 V	-2	-	2	-2	2	%
VTUNE_ADC	VTUNE ADC monitor range		0	-	3.3	0	3.3	V
	VTUNE ADC monitor effective resolution		-	3.32	-	-	-	mV
	VTUNE ADC monitor tolerance	VRECT = 3.0 V	-2	-	2	-2	2	%
IRECT_ADC	IRECT ADC monitor range		0	-	432	0	432	mA
	IRECT ADC monitor effective resolution		-	421.9	-	-	-	μA
	IRECT ADC monitor tolerance	IRECT = 200 mA	-7	-	7	-7	7	%
VOUT_ADC	VOUT ADC monitor range		0	-	5.4	0	5.4	V
	VOUT ADC monitor effective resolution		-	5.27	-	-	-	mV
	VOUT ADC monitor tolerance	VOUT = 4.2 V	-1	-	1	-1	1	%
IOUT_ADC	IOUT ADC monitor range		0	-	432	0	432	mA
	IOUT ADC monitor effective resolution		-	421.9	-	-	-	μA
	IOUT ADC monitor tolerance	IOUT = 200 mA	-6	-	6	-6	6	%

Table 45. General characteristics...continued

At recommended operating conditions; voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	T _{amb} = 25 °C			T _{amb} = -40 °C to +85 °C		Unit
			Min	Typ	Max	Min	Max	
NTC_ADC	NTC ADC monitor range		0	-	2.7	0	2.7	V
	NTC ADC monitor effective resolution		-	2.64	-	-	-	mV
	NTC ADC monitor tolerance	VNTC = 1.0V	-2	-	2	-2	2	%
TDIE_ADC	TDIE ADC monitor range		0	-	150	0	150	C
	TDIE ADC monitor effective resolution		-	5	-	-	-	C
INL	Integral Nonlinearity		-1.5	-	1.5	-1.5	1.5	LSB
DNL	Differential Nonlinearity		-1	-	1	-1	1	LSB
E _O	Offset error		-20	-	20	-20	20	mV
E _G	Gain error		-1	-	1	-1	1	%FS

Table 46. VPWR LDO characteristics

At recommended operating conditions; voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	T _{amb} = 25 °C			T _{amb} = -40 °C to +85 °C		Unit
			Min	Typ	Max	Min	Max	
V _{PWR}	VPWR LDO Output voltage	I ² C Programmable	-	-	-	1.8	3.3	V
V _{PWR_AC}	VPWR LDO output voltage accuracy		-	-	-	-5.5	5.5	%
I _{O_MAX}	Maximum output current		50	-	-	50		mA
V _{PWR_DO}	VPWR LDO drop out voltage	V _O = 3.3 V; I _O = 50 mA	-	150	-	-	250	mV
$\Delta V_{PWR} / \Delta V_I$	DC Line regulation	V _I = 5 V to 10 V; I _O = 50 mA	-	0.1	-	-	-	%/V
$\Delta V_{PWR} / \Delta I_O$	DC Load regulation	0 mA < I _O < 50 mA	-	-	-	0	1	%
PSRR	Power supply rejection rate	V _I = 5 V; I _O = 50 mA; fripple = 1 kHz; C _{VPWR} = 2.2 µF	-	50	-	-	-	dB
I _{OC_PWR}	VPWR LDO over current limit		-	55	-	50	-	mA
I _{SC_PWR}	VPWR LDO short circuit protection		-	150	-	100	-	mA

Table 47. VOUT Battery Charger characteristics

At recommended operating conditions; voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	T _{amb} = 25 °C			T _{amb} = -40 °C to +85 °C		Unit
			Min	Typ	Max	Min	Max	
I _{BAT_DISABLE}	VBAT quiescent Current VBAT = 4.5V AC = open, charger disabled		-	200	-	-	300	nA
BATTERY FAST-CHARGE MODE (CONSTANT CURRENT OR CONSTANT VOLTAGE)								
V _{BAT_REG}	V _{BAT_REG} typical range	I ² C Programmable; 20 mV/step	-	-	-	3.6	4.6	V
V _{BAT_REG_AC}	Constant Voltage Mode Regulation Voltage Accuracy	I _O = 0 mA to 200 mA; V _{RECTI} = 5.0 V, T _{amb} = 25 °C	-0.5		0.5	-1	1	%
ΔV _{BAT_REG} (HOT)			-	140	-	120	160	mV
I _{CHG_CC}	Constant charging current value	I ² C programmable, V _{BAT_REG} > V _{BAT_LOW} ; V _I = 5V	-	-	-	5	315	mA
I _{CHG_CC_AC}	Constant charging current accuracy	I _{CHG_CC} > 40 mA	-	-	-	-5	5	%
		20 mA < I _{CHG_CC} < 40 mA	-	-	-	-7.5	7.5	%
		5 mA < I _{CHG_CC} < 20 mA	-	-	-	-10	10	%
t _{CC_STEP_RISE}	Constant charging current ramp up step timing	5 mA/step	-	1	-	-	-	ms
I _{CC_FALL_SLEWRATE}	Constant charging current ramp down slew rate	When the charge from CC to IDLE	-	0.6	-	-	-	mA/μs
BATTERY PRE-CHARGE MODE (Dead battery and low battery)								
V _{BAT_DEAD}	Dead battery charge to low battery charge transition threshold		-	1.9	-	-	-	V
V _{BAT_DEAD_AC}	Dead battery charge to low battery charge transition threshold accuracy		-	-	-	-3.5	3.5	%
t _{DGL} (BAT_DEAD2LOW)	De-glitch time from dead battery charge to low battery charge transition		-	50	-	-	-	μs
I _{CHG_DEAD}	Dead battery charging current value	I ² C programmable, 1 mA/Step	-	-	-	1	63	mA
I _{CHG_DEAD_AC}	Dead battery charging current accuracy	V _{BAT} > 0.5V	-	-	-	-8	8	%
t _{DGL} (BAT_LOW2DEAD)	De glitch time from low battery charge to dead battery charge transition		-	50	-	-	-	μs
V _{BAT_LOW}	Pre-charge to fast charge transition threshold		-	3.0	-	-	-	V
V _{BAT_LOW_AC}	Pre-charge to fast charge transition threshold accuracy		-	-	-	-3.5	3.5	%

Table 47. VOUT Battery Charger characteristics...continued

At recommended operating conditions; voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	T _{amb} = 25 °C			T _{amb} = -40 °C to +85 °C		Unit
			Min	Typ	Max	Min	Max	
t _{DGL} (BAT_LOW2CC)	De-glitch time from low battery charge to fast charge transition		-	50	-	-	-	µs
I _{CHG_LOW}	Low battery charging current value	I ² C programmable, 1 mA/Step	-	-	-	1	63	mA
I _{CHG_LOW_AC}	Low battery charging current accuracy		-	-	-	-8	8	%
t _{DGL} (BAT_CC2LOW)	De-glitch time from fast charge to low battery charge transition		-	50	-	-	-	ms
BATTERY Top_off MODE (Top off mode)								
I _{CHG_TOPOFF}	Top off charging current value	I ² C programmable, 1 mA/Step	-	-	-	1	63	mA
I _{CHG_TOPOFF_AC}	Top off charging current accuracy	I _O ≥ 4 mA	-15	-	15	-17	17	%
t _{DGL} (BAT_CC2TOPOFF)	TOPOFF detection deglitch time on fast charging to top-off charging transition		-	22	-	-	-	ms
Battery Charger Restart								
V _{BAT_RESTART}	Charging restart threshold voltage	When V _{BAT} below V _{BAT_REG}	-	140	-	105	175	mV
t _{DGL} (BAT_RESTART)	De-glitch time, recharge threshold detected		-	50	-	-	-	ms
Battery Attach Detection								
I _{BAT_DET_SINK}	Sink current during battery detection	VRECT = 5 V	-	5	-	-	-	mA
t _{DGL} (BAT_DET_SINK)	De-glitch time, for sinking current	VRECT = 5 V	-	300	-	-	-	ms
t _{DGL} (BAT_DET_SOURCE)	De-glitch time, for source current	VRECT = 5 V	-	300	-	-	-	ms
V _{BAT_DET_LOW}	Battery detection lower threshold	VRECT = 5 V	-	1.9	-	-	-	V
V _{BAT_DET_UPPER}	Battery detection upper threshold	VRECT = 5 V	-	3.0	-	-	-	V
BATTERY-PACK NTC MONITOR(TS)								
I _{NTC_10K}	NTC thermistor bias current for 10k	V _{TS} < 2.7 V	-	50	-	48	52	µA
I _{NTC_100K}	NTC thermistor bias current for 100k	V _{TS} < 2.7 V	-	5	-	4.7	5.3	µA
V _{NTC(0°C)}	NTC threshold voltage at 0 °C	NTC_BETA_SEL[2:0] = 000	-	1.372	-	1.252	1.492	V
		NTC_BETA_SEL[2:0] = 001	-	1.462	-	1.342	1.582	V
		NTC_BETA_SEL[2:0] = 010	-	1.617	-	1.497	1.737	V
		NTC_BETA_SEL[2:0] = 011	-	1.614	-	1.494	1.734	V

Table 47. VOUT Battery Charger characteristics...continued

At recommended operating conditions; voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	T _{amb} = 25 °C			T _{amb} = -40 °C to +85 °C		Unit
			Min	Typ	Max	Min	Max	
		NTC_BETA_SEL[2:0] = 100	-	1.702	-	1.582	1.822	V
		NTC_BETA_SEL[2:0] = 101	-	1.780	-	1.66	1.900	V
		NTC_BETA_SEL[2:0] = 110	-	1.892	-	1.772	2.012	V
		NTC_BETA_SEL[2:0] = 111	-	2.025	-	1.905	2.145	V
V _{NTC(0°C)_HYS}	NTC threshold hysteresis voltage at 0 °C		-	110	-	-	-	mV
V _{NTC(10°C)}	NTC threshold voltage at 10 °C	NTC_BETA_SEL[2:0] = 000	-	0.900	-	0.820	0.980	V
		NTC_BETA_SEL[2:0] = 001	-	0.933	-	0.853	1.013	V
		NTC_BETA_SEL[2:0] = 010	-	0.989	-	0.909	1.069	V
		NTC_BETA_SEL[2:0] = 011	-	0.989	-	0.909	1.069	V
		NTC_BETA_SEL[2:0] = 100	-	1.018	-	0.938	1.098	V
		NTC_BETA_SEL[2:0] = 101	-	1.047	-	0.967	1.127	V
		NTC_BETA_SEL[2:0] = 110	-	1.087	-	1.007	1.167	V
		NTC_BETA_SEL[2:0] = 111	-	1.131	-	1.051	1.211	V
V _{NTC(10°C)_HYS}	NTC threshold hysteresis voltage at 10 °C		-	70	-	-	-	mV
V _{NTC(45°C)}	NTC threshold voltage at 45 °C	NTC_BETA_SEL[2:0] = 000	-	0.246	-	0.222	0.270	V
		NTC_BETA_SEL[2:0] = 001	-	0.236	-	0.208	0.260	V
		NTC_BETA_SEL[2:0] = 010	-	0.220	-	0.196	0.244	V
		NTC_BETA_SEL[2:0] = 011	-	0.220	-	0.196	0.244	V
		NTC_BETA_SEL[2:0] = 100	-	0.213	-	0.185	0.237	V
		NTC_BETA_SEL[2:0] = 101	-	0.205	-	0.181	0.229	V
		NTC_BETA_SEL[2:0] = 110	-	0.195	-	0.171	0.219	V
		NTC_BETA_SEL[2:0] = 111	-	0.186	-	0.160	0.210	V
V _{NTC(45°C)_HYS}	NTC threshold hysteresis voltage at 45 °C		-	24	-	-	-	mV
V _{NTC(60°C)}	NTC threshold voltage at 60 °C	NTC_BETA_SEL[2:0] = 000	-	0.151	-	0.139	0.168	V
		NTC_BETA_SEL[2:0] = 001	-	0.141	-	0.127	0.156	V
		NTC_BETA_SEL[2:0] = 010	-	0.126	-	0.114	0.144	V
		NTC_BETA_SEL[2:0] = 011	-	0.126	-	0.114	0.142	V
		NTC_BETA_SEL[2:0] = 100	-	0.119	-	0.102	0.131	V
		NTC_BETA_SEL[2:0] = 101	-	0.111	-	0.099	0.129	V
		NTC_BETA_SEL[2:0] = 110	-	0.102	-	0.089	0.116	V
		NTC_BETA_SEL[2:0] = 111	-	0.094	-	0.077	0.106	V
V _{NTC(60°C)_HYS}	NTC threshold hysteresis voltage at 60 °C		-	20	-	-	-	mV

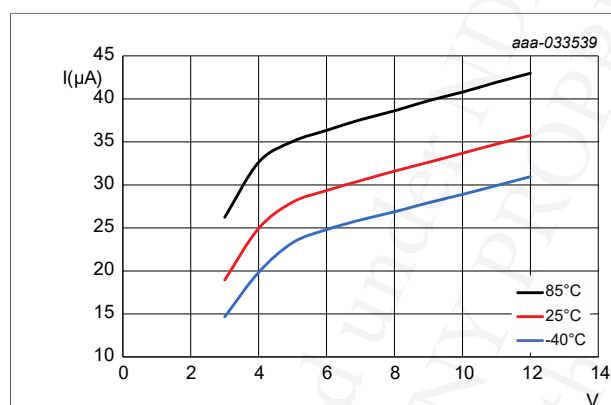
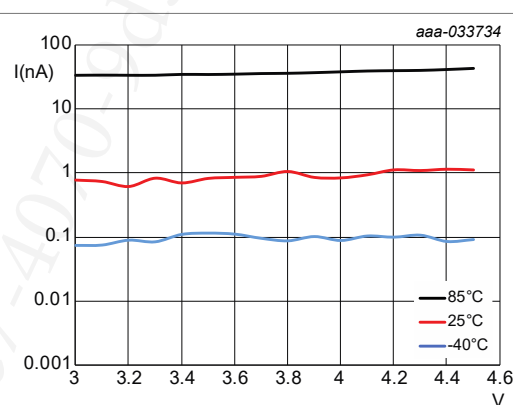
Table 47. VOUT Battery Charger characteristics...continued

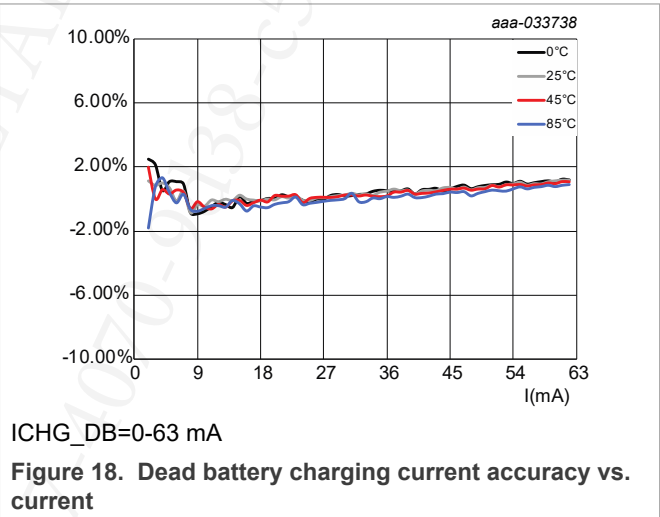
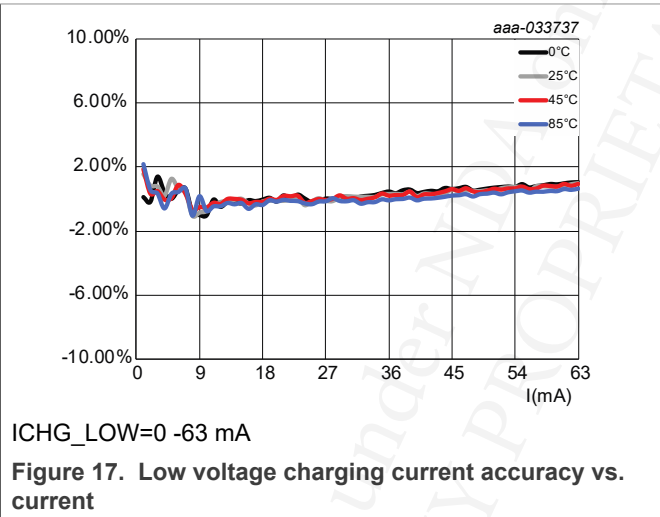
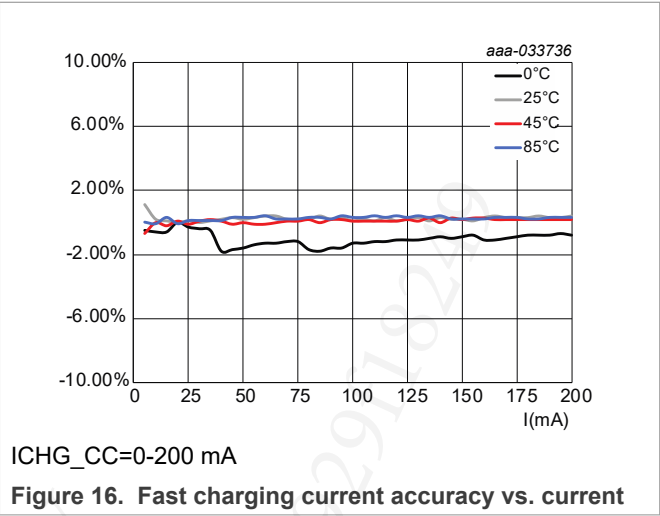
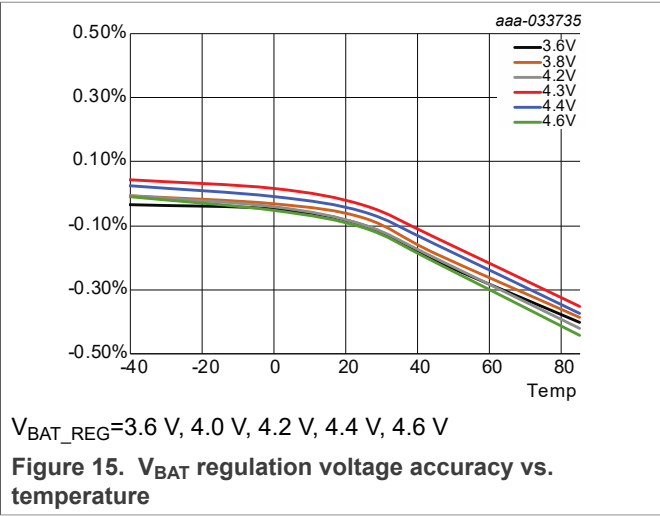
At recommended operating conditions; voltages are referenced to GND (ground = 0 V).

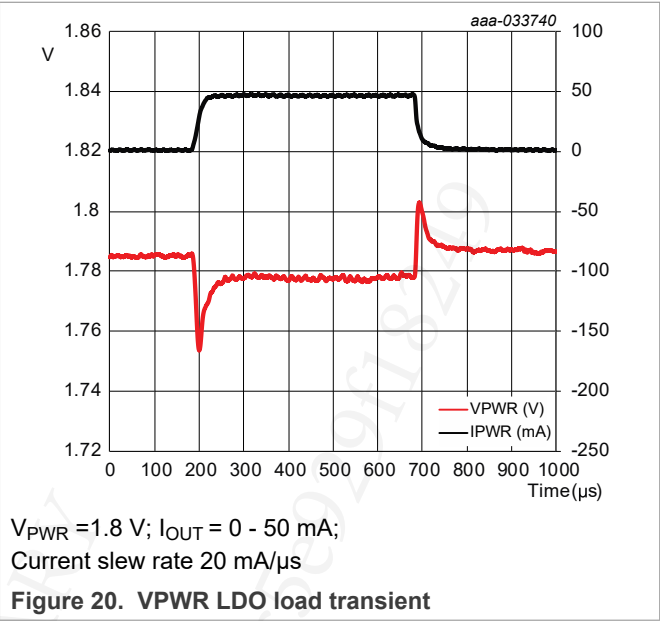
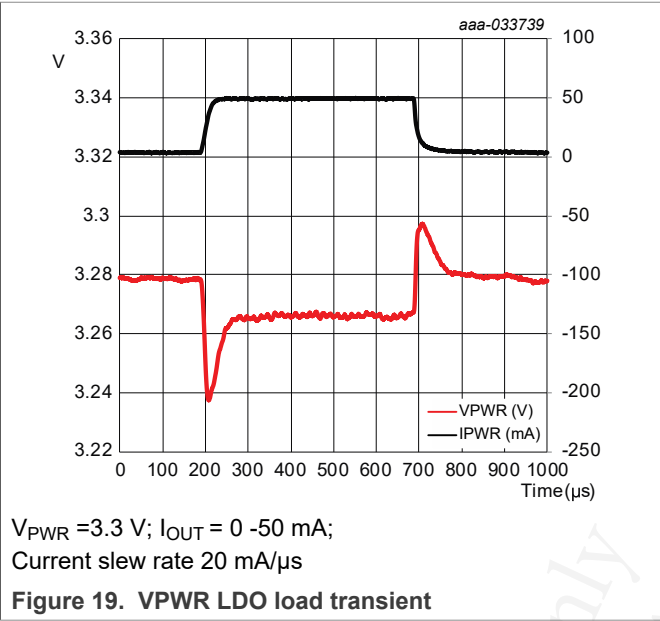
Symbol	Parameter	Conditions	T _{amb} = 25 °C			T _{amb} = -40 °C to +85 °C		Unit
			Min	Typ	Max	Min	Max	
t _{DGL} (NTC)	De-glitch time for NTC pin		-	50	-	-	-	ms
C _{NTC}	Maximum Decoupling Capacitor		-	10	-	-	-	nF
THERMAL REGULATION								
T _{J(REG)}	Thermal regulation (fold-back) range	I ² C Programmable, 5 °C/step	-	-	-	90	125	°C
T _{J(REG)_HYS}	Thermal regulation (fold-back) Hysteresis		-	20	-	-	-	°C
T _{J(OFF)}	Thermal shutdown temperature	When TEMP_SHDN[2:0] = 1'b100	-	140	-	-	-	°C
T _{J(OFF)_HYS}	Thermal shutdown temperature Hysteresis		-	20	-	-	-	°C

13.1 Graphs

The following graphs show the device performance of PCA9430.

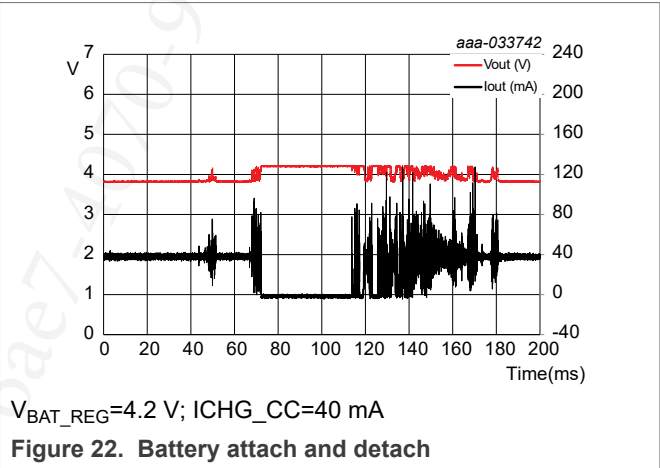
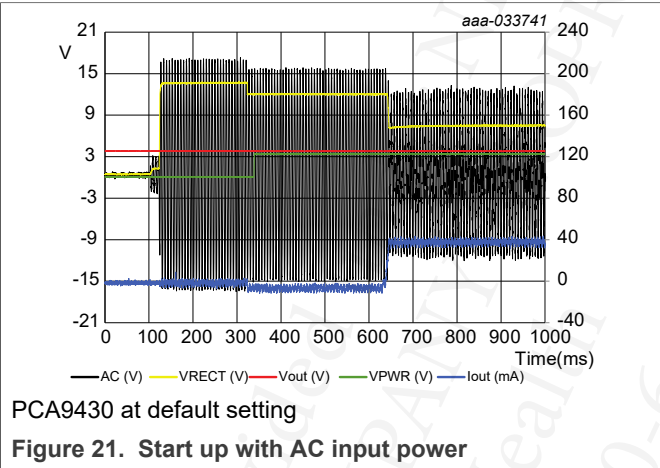
**Figure 13. SLEEP mode quiescent current****Figure 14. IBAT disable current vs. temperature**





13.2 System testing setup and waveforms

The PN7362 poller board and PCA9430 listener board are used for system testing. The TX and RX matching are optimized for 50 mA charging load.



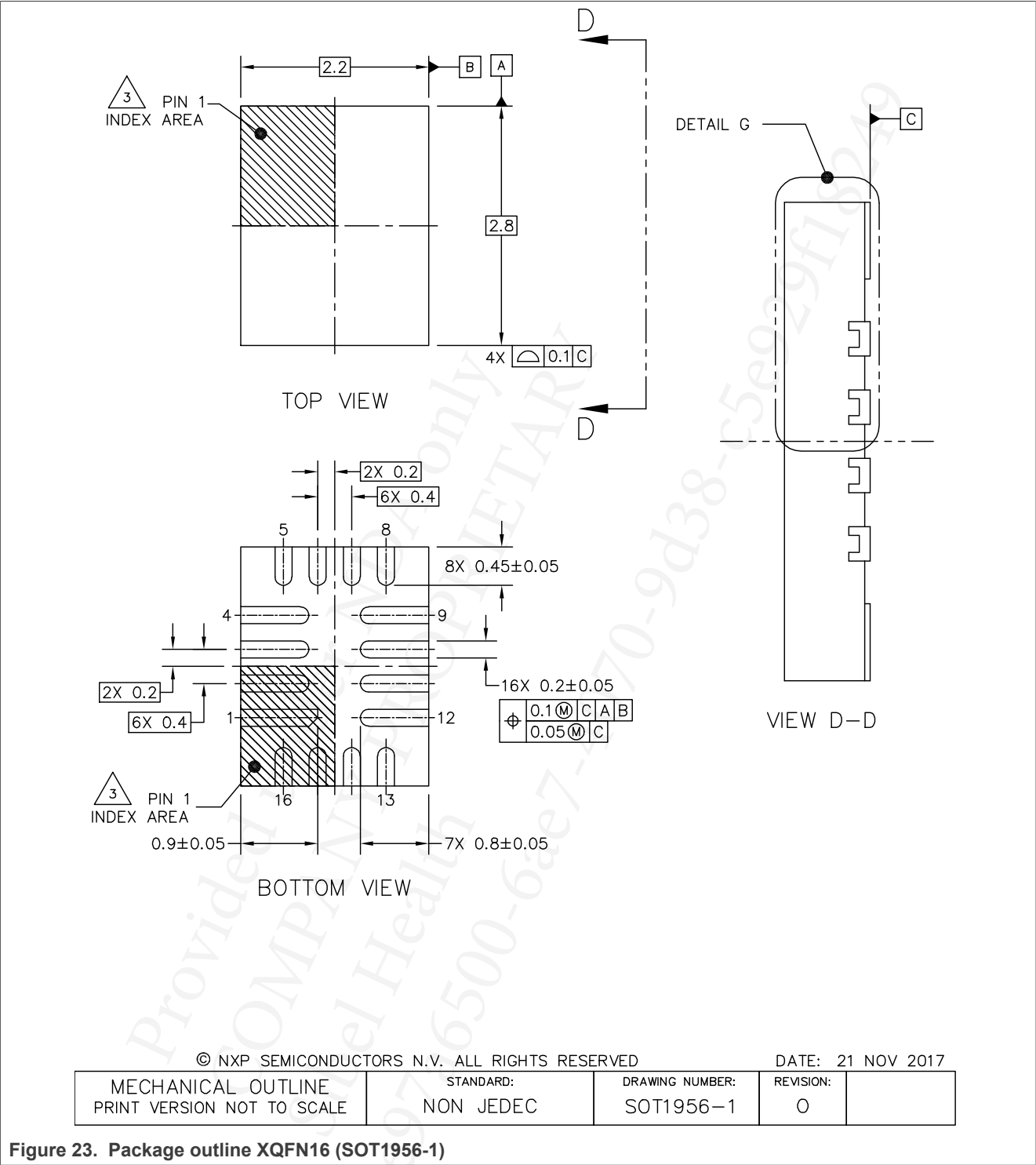
14 Dynamic characteristics

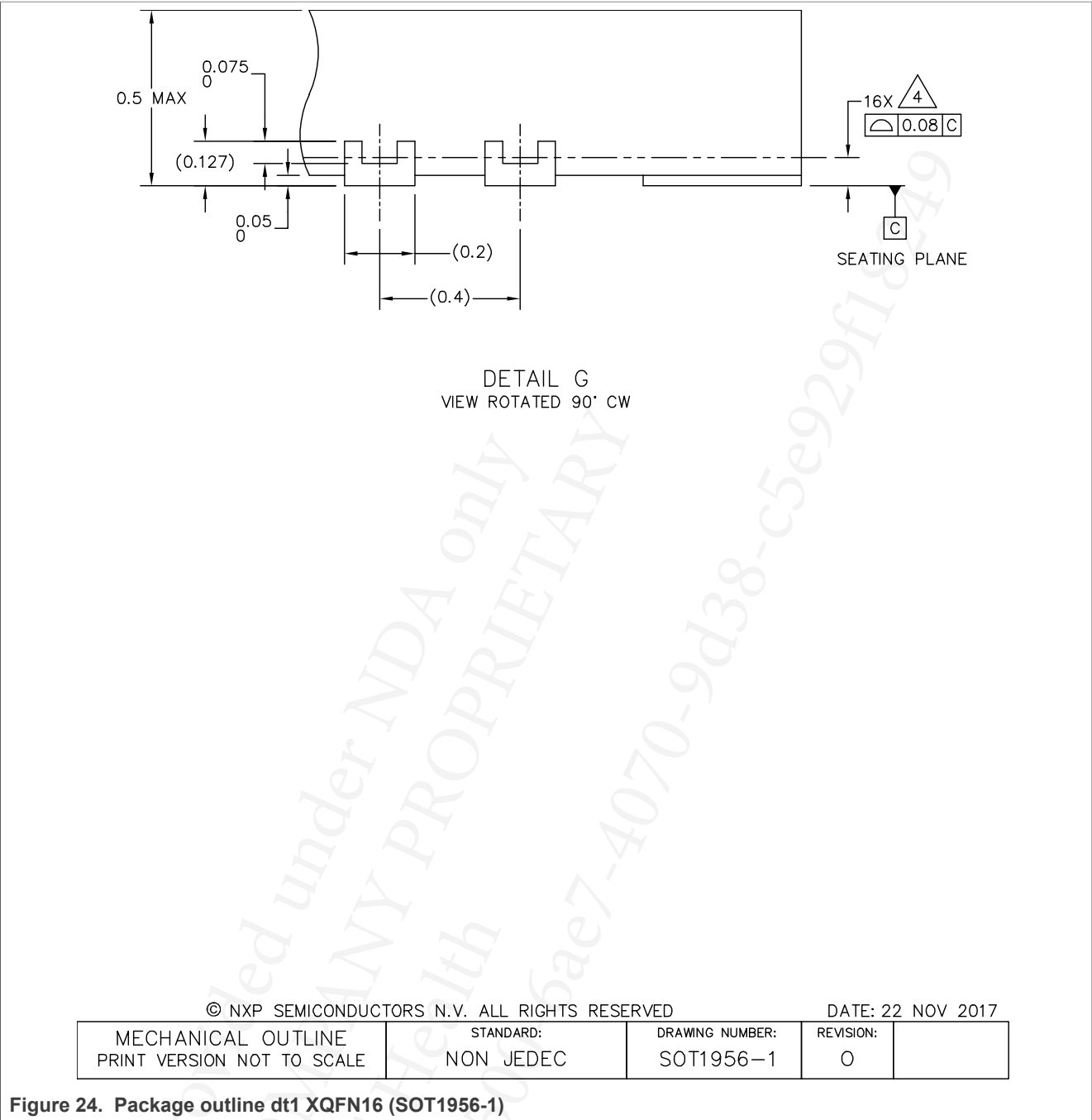
Table 48. I²C-bus interface timing requirements

At recommended operating conditions; $T_{amb} = -40\text{ }^{\circ}\text{C}$ to $+85\text{ }^{\circ}\text{C}$; voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	Standard Mode		Fast Mode		Fast Plus Mode		Unit
			Min	Max	Min	Max	Min	Max	
f_{SCL}	I ² C SCL clock frequency		0	100	0	400	0	1000	kHz
t_{HIGH}	HIGH period of the SCL clock		4.0	-	0.6	-	0.26	-	μs
t_{LOW}	LOW period of the SCL clock		4.7	-	1.3	-	0.5	-	μs
t_{SP}	pulse width of spikes that are suppressed by the input filter		-	50	-	50	-	50	ns
$t_{SU,DAT}$	data set-up time		250	-	100	-	50	-	ns
$t_{HD,DAT}$	data hold time		0	-	0	-	0	-	ns
t_r	rise time of both SDA and SCL signals		-	1000	20	300	-	120	ns
t_f	fall time of both SDA and SCL signals		-	300	$20 \times (V_{DD} / 5.5 \text{ V})$	300	$20 \times (V_{DD} / 5.5 \text{ V})$	120	ns
t_{BUF}	bus free time between a STOP and START condition		4.7	-	1.3	-	0.5	-	μs
$t_{SU,STA}$	set-up time for a repeated START condition		4.7	-	0.6	-	0.26	-	μs
$t_{HD,STA}$	hold time (repeated) START condition		4.0	-	0.6	-	0.26	-	μs
$t_{SU,STO}$	set-up time for STOP condition		4.0	-	0.6	-	0.26	-	μs
C_b	capacitive load for each bus line		-	400	-	400	-	550	pF
$t_{VD,DAT}$	data valid time	SCL LOW to SDA output valid	-	3.45	-	0.9	-	0.45	μs
$t_{VD,ACK}$	data valid acknowledge time	ACK signal from SCL LOW to SDA LOW	-	3.45	-	0.9	-	0.45	μs

15 Package outline





NOTES:

- 1. ALL DIMENSIONS ARE IN MILLIMETERS.
- 2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M–1994.
- 3. PIN 1 FEATURE SHAPE, SIZE AND LOCATION MAY VARY.
- 4. COPLANARITY APPLIES TO LEADS.

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MECHANICAL OUTLINE PRINT VERSION NOT TO SCALE	STANDARD: NON JEDEC	DRAWING NUMBER: SOT1956–1	REVISION: 0	

Figure 25. Package outline note XQFN16 (SOT1956-1)

16 Packing information

16.1 SOT1956-1; XQFN16; reel pack, SMD, 7" Q1 standard product orientation ordering code (12NC) ending 471

16.1.1 Dimensions and quantities

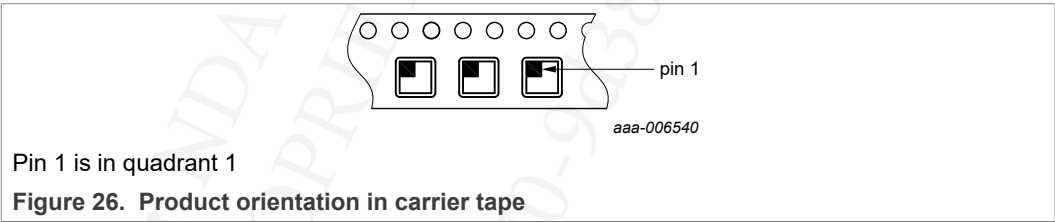
Table 49. Dimensions and quantities

Reel dimensions d × w (mm) ^[1]	SPQ/PQ (pcs) ^[2]	Reels per box
180 x 12	4000	1

[1] d = reel diameter; w = tape width.
[2] Packing quantity dependent on specific product type.

View ordering and availability details at [NXP order portal](#), or contact your local NXP representative.

16.1.2 Product orientation



16.1.3 Carrier tape dimensions

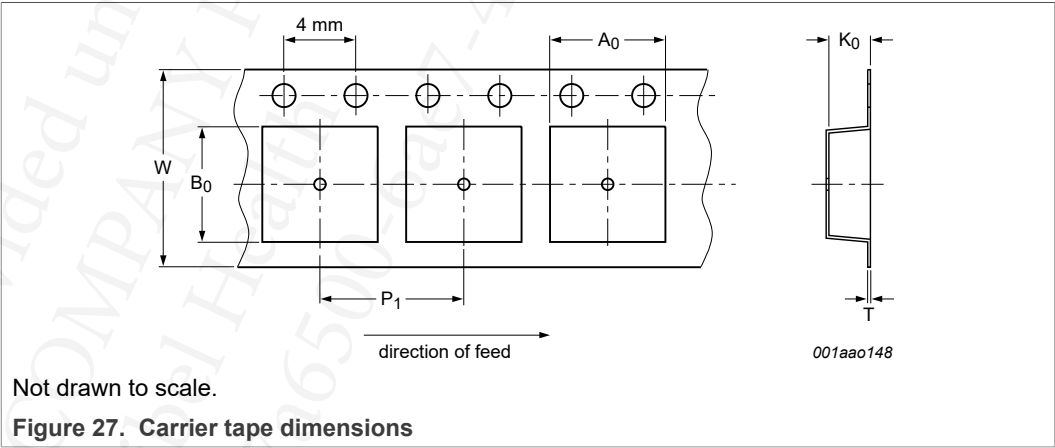
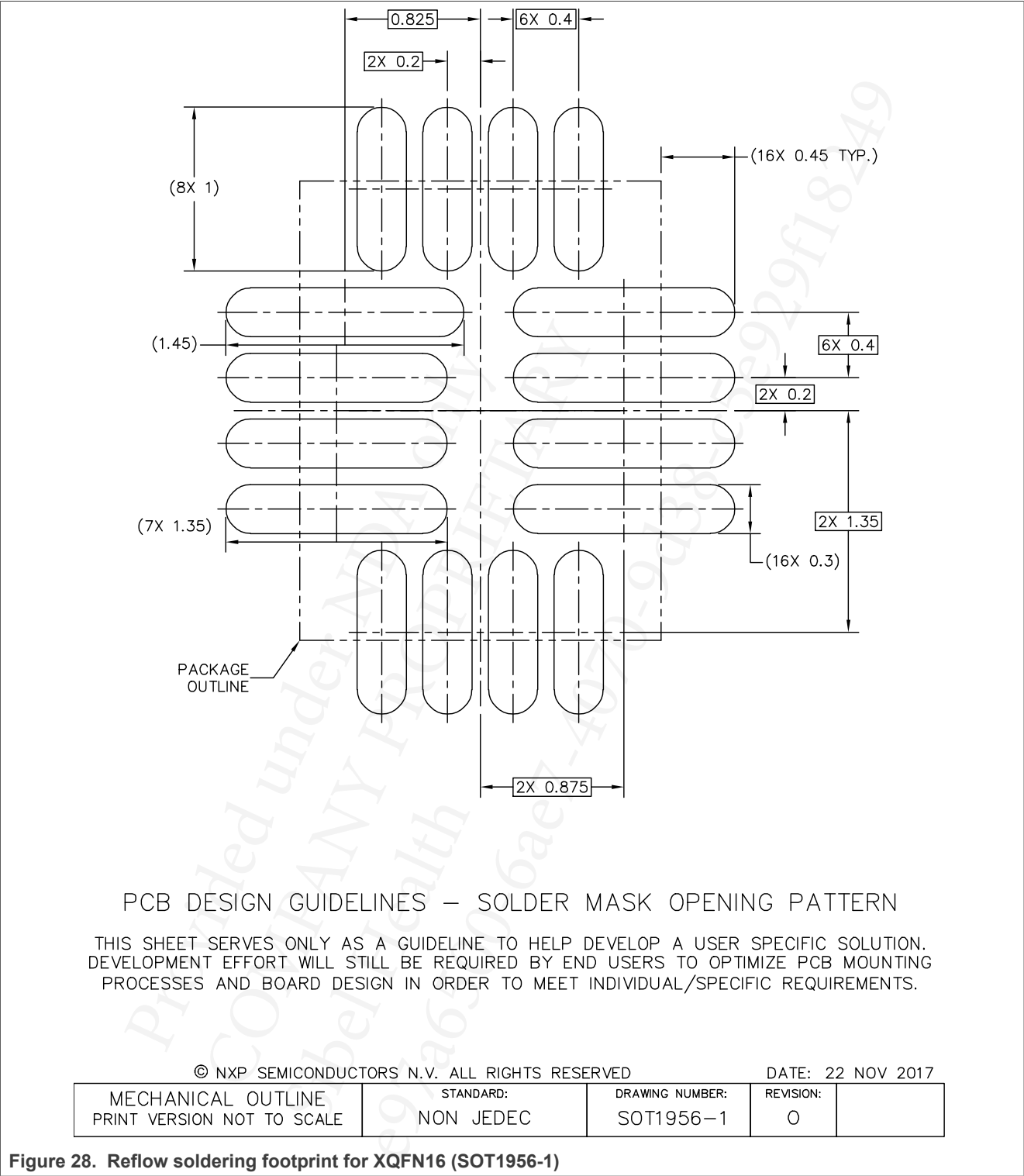


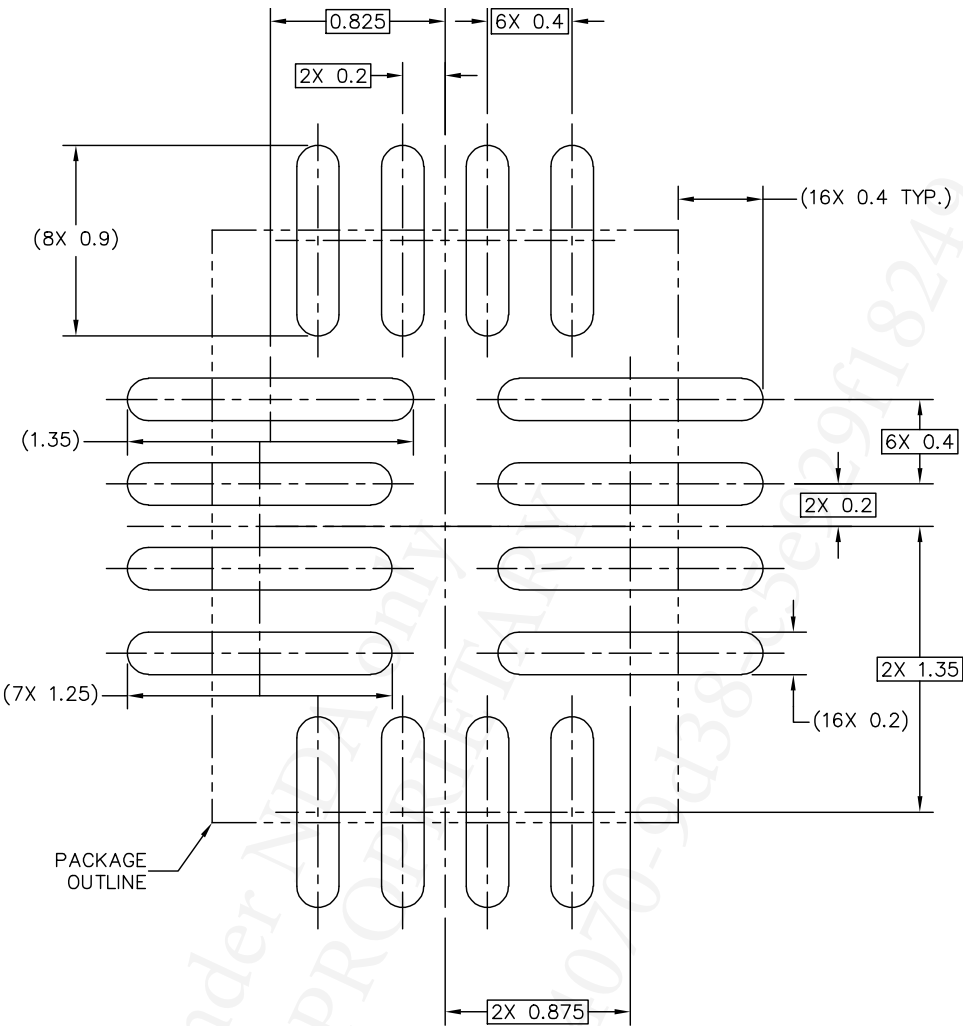
Table 50. Carrier tape dimensions

In accordance with IEC 60286-3/EIA-481; this pack method utilizes an 'SBB' (Sulfur Barrier Bag) solution.

A ₀ (mm)	B ₀ (mm)	K ₀ (mm)	T (mm)	P ₁ (mm)	W (mm)
2.40 ± 0.05	3.00 ± 0.05	0.70 ± 0.10	0.25 ± 0.03	4.0 ± 0.1	12 +0.3/-0.1

17 Soldering





PCB DESIGN GUIDELINES – I/O PADS AND SOLDERABLE AREA

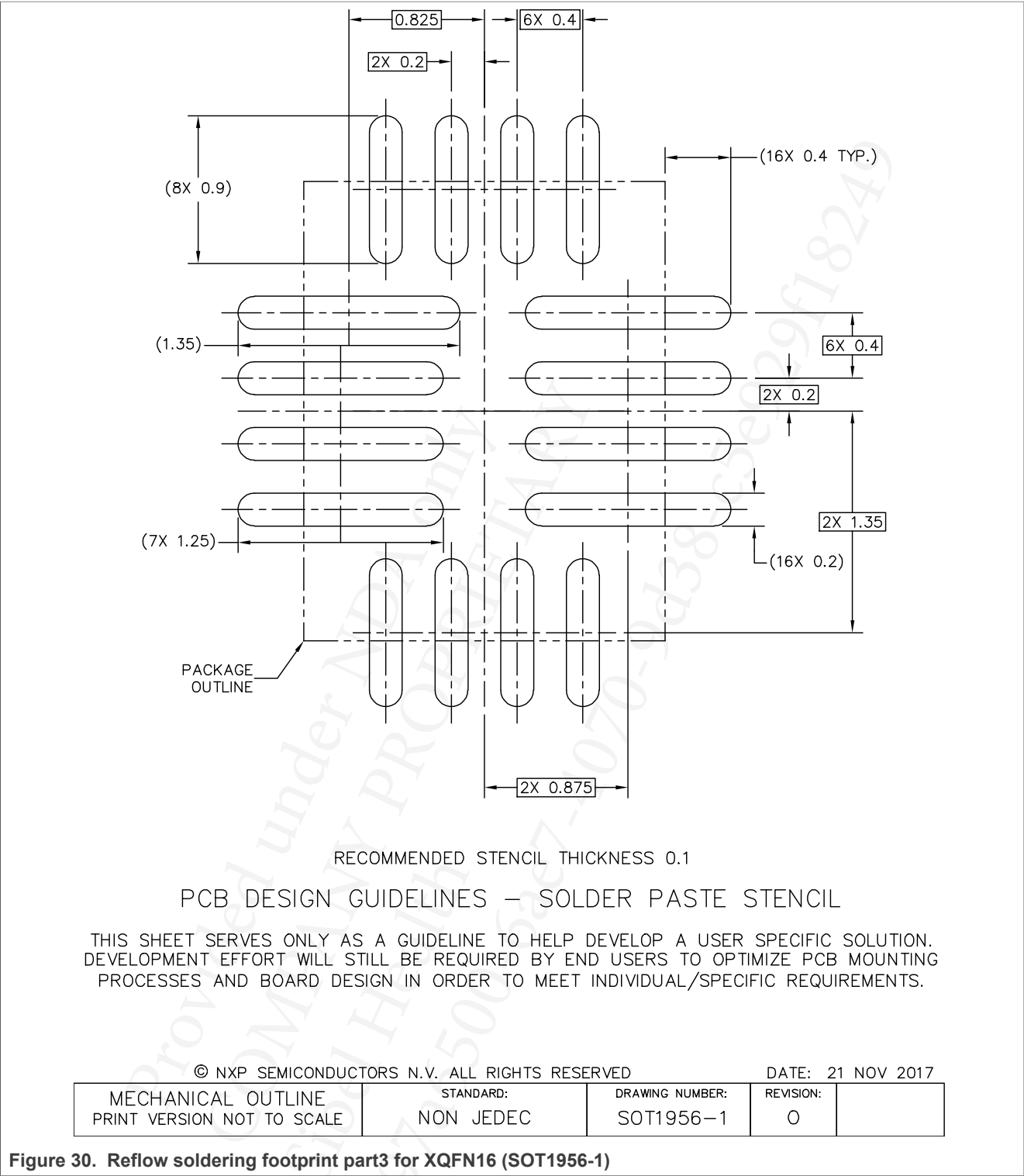
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DATE: 22 NOV 2017

MECHANICAL OUTLINE PRINT VERSION NOT TO SCALE	STANDARD: NON JEDEC	DRAWING NUMBER: SOT1956-1	REVISION: O	
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Figure 29. Reflow soldering footprint part2 for XQFN16 (SOT1956-1)



18 Abbreviations

Table 51. Abbreviations

Acronym	Description
ESD	ElectroStatic Discharge
CDM	Charged Device Model
HBM	Human Body Model
NFC	Near Field Communication
ADC	Analog to Digital Converter
DAC	Digital to Analog Converter
WLC_L	Wireless charging listener
WLC_P	Wireless charging poller

19 Revision history

Table 52. Revision history

Document ID	Release date	Data sheet status	Change notice	Supersedes
PCA9430 v3.0	20201202	Product data sheet	-	PCA9430 v1.1
Modifications:	- Updated main title to "NFC charging power receiver with linear battery charger" Section 2: Updated internal rectifier to 200 mW - Updated Figure 1 Section 7.2: Updated VTUNE pin description Section 8.4: Added external rectifier information Section 8.9: Changed part from "NTAG I2C Plus" to "CRN120" Section 9: Split application diagram into two diagrams: internal rectifier and external rectifier Table 43: Split input current for rectifier into two conditions: internal rectifier and external rectifier			
PCA9430 v1.1	20190604	Product data sheet	-	-
Modifications:	- Updated P_{tot} TBD in Table 42 - Updated $I_{BAT_DISABLE}$ TBD in Table 47			
PCA9430 v1.0	20190514	Product data sheet	-	-

20 Legal information

20.1 Data sheet status

Document status ^{[1][2]}	Product status ^[3]	Definition
Objective [short] data sheet	Development	This document contains data from the objective specification for product development.
Preliminary [short] data sheet	Qualification	This document contains data from the preliminary specification.
Product [short] data sheet	Production	This document contains the product specification.

[1] Please consult the most recently issued document before initiating or completing a design.

[2] The term 'short data sheet' is explained in section "Definitions".

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